

DEPARTMENT OF INFORMATION COMMUNICATION TECHNOLOGY

Project Title: FinSmart AI :Personal Financial Assistant

Submitted in partial fulfillment of the requirements for the Bachelor of Technology in Information Communication Technology at RP Huye College

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July, 2025

DEPARTMENT OF INFORMATION COMMUNICATION TECHNOLOGY

DECLARATION(A)

I hereby declare that this project, titled "**FinSmart AI: Personal Financial Assistant,**" submitted in partial fulfillment of the requirements for the **Bachelor of Technology in Information Communication Technology** at **RP Huye College**, is my original work and has not been submitted, either in part or in full, for any other academic degree, diploma, or certification at any institution. A complete list of references is provided to acknowledge all sources of information used or cited in this work. This research adheres to institutional and ethical guidelines, including those related to data privacy, responsible system development, and academic integrity.

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DECLARATION(B)

This is to certify that the research project titled "**FinSmart AI: Personal Financial Assistant**" has been reviewed and approved for submission by **RP-Huye College**. The project was conducted under the supervision of **UMUTONI Marie** (Supervisor) and **MPUMUJE Maurice** (Co-Supervisor). It meets the academic standards and requirements for the **Bachelor of Technology in Information Communication Technology** program. It has been evaluated for its originality, methodological soundness, and relevance to the field, and is considered suitable for academic recognition.

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AUTHORITY TO DEPOSIT THE CORRECTED VERSION OF THE PROJECT TO THE LIBRARY

I the undersigned hereby do testify to have verified the corrections made by the student Elyse TUMUSHIME.

To their Project entitled " **FinSmart AI: Personal Financial Assistant** " and authorized her to deposit the document to the library of the RP Huye College.

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DEDICATION

I dedicate this work to the Almighty God for His mercy and love.

To my parents, for their endless support and encouragement.

To all of my family, for their constant love and care.

To my supervisor, your insights have proven priceless on this adventure and to everyone who contributed to the completion of this work.

Finally, I dedicate this work to all my colleagues, you've shaped my world this past year in ways I can't express.

ACKNOWLEDGMENTS

First and foremost, I sincerely thank **God Almighty** for granting me the strength, wisdom, and perseverance to complete this project. Without His continuous guidance and grace, this achievement would not have been possible. I extend my heartfelt gratitude to my supervisor, **Ms. Marie UMUTONI** and my Co-supervisor **MPUMUJE Maurice** , whose unwavering support, constructive feedback, and expert guidance have been instrumental in shaping the direction and quality of this work. I am also deeply thankful to the **ICT Department** at **RP-Huye College**, especially **Mr. NDIZEYE Patrick**, for always being available with advice, resources, and encouragement. To all my lecturers, your dedication and mentorship have helped me grow academically and personally throughout this program. I am immensely grateful to my family, including my parents, siblings, and loved ones, for their constant support, belief in me, and encouragement during every step of this journey. Your love and motivation have been my driving force. A heartfelt thank you goes to my **friends, classmates, and colleagues** for your collaboration, team spirit, and all the late-night discussions that made this process more enjoyable and manageable. Finally, to everyone who contributed to this project in any way, thank you. This journey has been filled with learning, challenges, and personal growth, and I truly appreciate each and every one of you.

ABSTRACT

Financial management is challenging, especially for low-income earners relying on cash-based transactions and facing unexpected expenses. Without proper budget tools, tracking income, controlling spending, and saving money effectively become difficult, leading to financial instability. This study develops **FinSmart AI, an accessible personal financial assistant** that leverages OpenAI-powered natural language processing (NLP) and machine learning (ML) to analyze financial and linguistic features such as monthly income, total expenses, savings goals, money-related terms, and negations. Two ML models—**Random Forest** and **Decision Tree** classifiers—were trained, achieving **100%** training and test accuracy for Random Forest and **94.44%** test accuracy for Decision Tree. The Random Forest model was selected to predict whether users will meet their financial goals or not, enabling personalized financial guidance. Implemented via a Flask-based web interface, FinSmart AI offers real-time chatbot interaction, automated email alerts for income or budget changes, and spending habit summaries with tailored tips on login. In emergencies, the system suggests a 20% reduction in the highest spending categories to preserve essential needs. An administrative dashboard enables user and behavior management. This research demonstrates that integrating interpretable ML models, OpenAI NLP, and automated communication in an online platform can deliver scalable, user-friendly financial management tools that empower low-income users to improve spending habits, achieve goals, and foster long-term financial stability. To increase accessibility and encourage daily use, this study recommends supporting Kinyarwanda localization, integrating mobile accessibility, and promoting FinSmart AI as a personal finance companion for everyday financial decisions.

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LIST OF SYMBOLS AND ABBREVIATIONS/ACRONYMS

- ❖ **AI** : Artificial Intelligence
- ❖ **FinSmart** : Financial Smart
- ❖ **NLP**: Natural Language Processing
- ❖ **ML**: Machine Learning
- ❖ **TAM**: Technology Acceptance Model
- ❖ **SPSS** :Statistical Package for the Social Sciences
- ❖ **SDLC**: Software Development Life Cycle
- ❖ **UI**: Portable Document Format
- ❖ **XAMPP**: Cross-platform Apache, MySQL, PHP, and Perl
- ❖ **SQL**: Structured Query Language
- ❖ **HTML** :HyperText Markup Language
- ❖ **CSS** : Cascading Style Sheets
- ❖ **PHP** : Hypertext Preprocessor
- ❖ **OPENAI**: Open Artificial Intelligence
- ❖ **IPRC**: Integrated Polytechnic Regional College
- ❖ **RP**: Rwanda Polytechnic

DEFINITION OF KEY TERMS

1. **Artificial Intelligence (AI):**

A field of computer science where systems mimic human intelligence in tasks like learning and decision-making. FinSmart AI uses AI to offer personalized financial advice via chatbot.

2. **Machine Learning (ML):**

A branch of AI where systems learn from data to make predictions. FinSmart AI uses ML to analyze user behavior and predict goal outcomes.

3. **Financial Literacy:**

The ability to understand budgeting, saving, and spending. FinSmart AI promotes this by providing tips based on user habits.

4. **Budgeting:**

The process of planning income and expenses. FinSmart AI automatically creates and adjusts budgets based on income, spending, and emergencies.

5. **Cash-Based Transactions:**

Payments made using physical money, often used by low-income earners. The system allows manual tracking to improve budgeting.

6. **Random Forest:**

An ensemble ML algorithm using multiple decision trees to boost accuracy. In this project, it achieved **100% test accuracy**.

7. **Decision Tree:**

An ML model that predicts outcomes based on input rules. It was used for comparison and reached **94.44% test accuracy**.

8. **Natural Language Processing (NLP):**

An AI field focused on understanding human language. Enables FinSmart AI to interpret user messages and give contextual responses.

9. **Email Notification System:**

An automated tool that sends users emails with financial summaries, spending insights, and budgeting advice.

10. **Emergency Budget Adjustment:**

A system feature that reallocates funds (e.g., reducing 20% from top categories) when emergencies arise, helping users avoid touching their savings.

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CHAPTER ONE: INTRODUCTION

INTRODUCTION

In today's rapidly evolving digital world, managing personal finances remains a major challenge for low-income individuals, especially those with limited access to digital tools and formal financial education. Traditional budgeting methods, such as handwritten ledgers and spreadsheets, are not only time-consuming but also prone to human error. These approaches often fail to support users in building long-term financial stability or preparing for unforeseen expenses. Meanwhile, the financial landscape continues to grow more complex, requiring adaptive and data-driven decision-making tools (Worldbank, 2022). Unfortunately, low-income earners are frequently excluded from such advancements due to infrastructural and educational barriers. The digital transformation of financial services has mainly prioritized affluent and digitally competent users, neglecting the needs of vulnerable populations (OECD., 2020). This lack of inclusivity in financial technology contributes to widening socioeconomic disparities. Individuals who are unable to access or use intelligent financial tools are more likely to experience budgeting challenges, overspending, and limited savings. Therefore, there is an urgent need for practical, user-oriented financial solutions that are both intelligent and accessible. Addressing this digital and financial divide is essential for promoting inclusive economic participation.

FinSmart AI is introduced as a purposeful response to these challenges, offering an intelligent financial assistant tailored to the unique needs of low-income users. By integrating Artificial Intelligence (AI) and Machine Learning (ML), the system provides dynamic budgeting support, spending summaries, and real-time alerts. Its goal is to improve financial awareness and enable users to make informed, confident decisions about their money. The platform is designed with simplicity and accessibility at its core, ensuring usability regardless of a person's educational or digital background. Unlike existing solutions that prioritize advanced features for high-income markets, FinSmart AI focuses on delivering essential financial insights in an understandable and personalized format (Nguyen, T., & Tran, L., 2018). It promotes financial inclusion by equipping users with tools to track spending patterns and develop positive habits over time. In doing so, the system contributes to reducing dependency on informal savings methods and improving overall economic resilience. Its adaptability also allows it to evolve with user behavior, offering ongoing value ((PricewaterhouseCoopers), 2017). This chapter

outlines the project's background, highlights the underlying problem, states the objectives and significance of the solution, and presents the structure of the remaining chapters.

1.1 BACKGROUND OF THE STUDY

Since 2015, Artificial Intelligence (AI) and Machine Learning (ML) technologies have rapidly advanced and transformed multiple sectors globally, including finance, healthcare, transportation, and education. By 2018, AI applications in finance had evolved to provide personalized budgeting support, spending analysis, and decision-making assistance based on user behavior (Davenport T. H., 2018). Despite these innovations, as of 2021, a significant portion of low-income individuals worldwide still face considerable challenges in managing their personal finances due to limited access to digital financial tools and low levels of financial literacy (Nguyen, 2021). Many continue to rely on traditional budgeting methods such as manual record-keeping or basic spreadsheets, which often lead to inaccurate expense tracking, poor savings, and inadequate preparation for emergencies.

At the national and local levels in Rwanda, these challenges are particularly pronounced. Although Rwanda has made significant progress in improving digital infrastructure and promoting financial inclusion, many low-income and rural communities still lack access to personalized financial guidance that supports informed decisions on saving and spending. Most digital financial platforms tend to serve urban, more educated, and higher-income users, leaving vulnerable populations underserved. This gap hinders Rwanda's goals for inclusive economic growth and poverty reduction. Researchers highlight the need for AI-powered solutions that provide accessible, context-aware financial guidance designed to empower low-income users to develop sustainable financial habits and improve their financial well-being (Worldbank, 2022). FinSmart AI aims to address this need by offering tailored, intelligent financial advice suited to the realities of Rwanda's underserved populations, helping them achieve greater financial stability and independence.

1.2 PROBLEM STATEMENT

Many low-income earners struggle with managing their personal finances due to reliance on manual expense tracking methods. These traditional approaches are often inaccurate and inconsistent, which makes budgeting difficult and prevents effective financial planning (Brynjolfsson E. &, 2014). This problem leads to overspending and irregular saving habits that increase financial vulnerability.

In addition, there is a lack of personalized financial guidance tailored to the income and spending behaviour of low-income individuals. Most existing solutions are designed for wealthier or more technologically experienced users, making it difficult for vulnerable populations to receive relevant advice that fits their financial realities (Davenport T. H., 2018). This gap limits their ability to prepare for unexpected expenses and build financial resilience.

Moreover, current digital financial tools often fail to simplify decision-making and encourage sustainable saving habits among underserved communities. Without intelligent, accessible, and context-aware support, many low-income earners remain financially exposed and unable to improve their long-term economic well-being (Choi J. , 2017)). Therefore, developing an AI-powered, user-friendly financial assistant like FinSmart AI is essential to bridge this gap and empower these individuals.

1.3 RESEARCH OBJECTIVES

1.3.1 General objectives

The main objective of the study is to develop an accessible personal finance assistant (FinSmart AI) that can help low-income earners to manage their finances efficiently. The system will provide real time expense tracking, individualized budgeting information and financial guidance to improve saving habits and prepare for unexpected expenses. Through simplified financial management, FinSmart AI aim to enhance financial literacy and foster long-term financial stability.

1.3.2 Specific objectives

In addition to the overarching goal, the specific objectives of FinSmart AI provide an easy-to-use financial management solution accessible to low-income earners through AI powered insights. The intended primary objectives of this study are:

- ❖ To develop a chatbot-based system that allows users to enter, categorize, and monitor income, expenses, and savings goals for seamless financial tracking.
- ❖ To provide AI-driven budgeting advice tailored to users' spending habits to support effective budget adherence.
- ❖ To implement contextual financial strategies that assist users in managing unexpected expenses without compromising essential needs.
- ❖ To generate personalized financial summaries with actionable recommendations to enhance users' saving habits.
- ❖ To improve system accessibility and usability by implementing a chatbot and automated email notifications that require no technical expertise or external installations.

1.4 HYPOTHESIS OF THE PROJECT

This study explores the impact of integrating a personalized AI-powered financial assistant into the daily money management practices of low-income earners in Rwanda. The research seeks to determine whether the system leads to measurable improvements in financial behavior, particularly in areas such as budgeting accuracy, saving discipline, and the ability to handle unexpected expenses.

Null Hypothesis (H_0):

Integrating a personalized AI chatbot into the financial routines of low-income earners in Rwanda has no statistically significant effect on improving their budgeting accuracy, saving habits, or preparedness for financial emergencies. Any observed changes in financial behavior are assumed to be due to chance or external factors unrelated to the system.

Alternative Hypothesis (H_1):

Integrating a personalized AI chatbot into the financial routines of low-income earners in Rwanda has a statistically significant positive impact on their budgeting accuracy, saving discipline, and readiness for unexpected expenses. The system contributes meaningfully to

better financial decision-making and overall financial well-being by offering real-time, tailored guidance and intelligent tracking tools.

1.5 SCOPE OF THE PROJECT

This project focuses on developing an AI-powered personal finance assistant, FinSmart AI, **geographically** tailored to Huye District in the Southern Province of Rwanda, with potential scalability to other districts facing similar challenges. The **theoretical scope** addresses limited financial literacy, reliance on manual budgeting, and the need for accessible personalized financial guidance using AI and natural language processing. The **content scope** includes income and budget categorization, expense tracking, emergency budget adjustment, and automated email notifications, while excluding advanced financial services like investment planning and credit scoring to maintain simplicity.

1.6 SIGNIFICANCE OF THE STUDY

1.6.1 Personal Interest

This project reflects my passion for developing innovative, practical solutions that create real-world impact. Through FinSmart AI, I seek to advance my expertise in artificial intelligence, machine learning, and software development, while directly addressing the financial challenges faced by low-income earners.

1.6.2 Common Interest

FinSmart AI tackles the critical issue of financial insecurity among vulnerable populations by delivering an accessible, AI-powered personal finance assistant. This initiative not only promotes better financial habits but also supports Rwanda's Vision 2050, highlighting financial literacy as a cornerstone for sustainable economic growth and community empowerment.

1.7 ORGANIZATION OF THE PROJECT

This project is subdivided into 5 chapters:

❖ **Chapter 1:** Introduction which introduces the Background of the study, statement of the problem, objectives of the project, scope of the project, significance of the project, and structure of the project.

❖ **Chapter 2:** the literature review; this part of the work discusses the key concepts with a description of different aspects relating to the research;

❖ **Chapter 3:** This will discuss how the data shall be presented, analyzed, and interpretation of results. The presentation format of data is designed based on the data collection instruments.

❖ **Chapter 4:** System Design, Implementation, and Testing

This chapter presents the design and implementation of FinSmart AI, followed by the results of system testing to evaluate its functionality and effectiveness.

❖ **Chapter 5:** Conclusion and Recommendation.

CHAPTER TWO: LITERATURE REVIEW

2.1 INTRODUCTION

This chapter presents a detailed review of the existing literature related to intelligent systems for personal finance management. It explores key technological advancements in artificial intelligence and machine learning that contribute to developing automated financial advisory tools. The chapter also examines expert insights, theoretical foundations, and previous studies that inform the design and implementation of AI-driven financial assistants. Through this review, the chapter establishes a foundation for understanding how intelligent systems can improve financial decision-making and literacy.

2.1 CONCEPTS, OPINIONS, AND IDEAS FROM AUTHORS/EXPERTS

This section explores the key scholarly ideas related to the study's core variables: AI-powered financial assistance, chatbot-based interaction, and behavioural change in financial habits. The reviewed literature aligns with the hypothesis that "Integrating a personalized AI chatbot into the financial routines of low-income earners in Rwanda has a statistically significant positive impact on their budgeting accuracy, saving discipline, and readiness for unexpected expenses." The literature is organized into sub-themes that mirror the research objectives and highlight both theoretical insights and gaps that the current study addresses.

2.1.1 concepts

Artificial Intelligence (AI) and machine learning (ML) are reshaping how individuals manage personal finances by enabling systems to offer predictive, personalized support. According to (Mitchell, 1997) machine learning identifies patterns in historical financial data, allowing systems to suggest savings goals or forecast spending trends. Chatbots, enhanced with natural language processing (NLP), serve as digital assistants that facilitate real-time financial conversations with users, improving both accessibility and engagement (Dix, 2004)

In the Rwandan context, however, financial management is still largely manual—using verbal tracking, handwritten notes, or informal savings cooperatives. According to the (FinScope Rwanda, 2020) survey, approximately 78% of adults in Rwanda use informal financial services, mainly through savings groups (ibimina). These traditional methods lack both automation and the intelligence needed to reinforce positive financial behaviour, making it difficult for individuals to build consistent saving habits or plan for emergencies. This reveals a disconnect between global technological advancements and local practices, highlighting a literature gap

in context-specific financial technology solutions for low-income earners in sub-Saharan Africa.

2.1.2 Opinions

However, most global financial technologies are not designed with the realities of informal economies in mind. According to the (FinScope Rwanda, 2020)report, many Rwandans continue to depend on informal savings methods and lack exposure to digital tools that can guide behaviour change. While previous studies ((Dix, 2004); (Harlow, 2008)) emphasized that contextual fit and behavioural alignment are critical for sustained engagement, few have explored how these principles apply in sub-Saharan African settings where financial behaviour is culturally rooted and largely undocumented. This reveals a critical gap in literature: a lack of AI-powered financial tools tailored to informal economies and behavioural patterns found in low-income communities. This study addresses that gap by designing and testing a personalized AI chatbot that not only delivers financial guidance but is also aligned with the saving habits, language preferences, and budget patterns of Rwandan low-income earners—thereby enabling improved budgeting accuracy, saving discipline, and financial resilience.

2.1.3 Ideas from Authors and Experts

Several scholars emphasize the role of digital alerts as effective tools in promoting positive financial behaviour. According to (Thaler, 2008)timely and clear notifications can significantly improve users’ attention to their financial goals by providing immediate feedback and reminders. In the context of personal finance management, alerts serve as active prompts that help users stay aware of their spending limits and encourage saving habits (Kim, 2020). These alerts enhance user engagement by delivering relevant information exactly when it is needed, thus supporting behaviour change.

However, experts also stress that alerts must be context-aware and culturally relevant to be effective. (Dix, 2004)) argues that technology adoption in low-income and informal economies depends on how well digital systems fit users’ routines and financial realities. In Rwanda, where informal saving and spending patterns dominate, alerts need to be tailored to local financial behaviours and communication preferences. This highlights a gap in current literature, which mostly focuses on formal economies and digital literacies common in developed countries. This study aims to address this by designing an AI-powered alert system specifically adapted for low-income earners in Rwanda, improving budgeting accuracy, saving discipline, and readiness for emergencies.

2.2 ARTIFICIAL INTELLIGENCE IN PERSONAL FINANCE

Artificial Intelligence (AI) refers to the simulation of human intelligence processes by machines, especially computer systems. In personal finance, AI enables automation, personalized advice, and predictive analytics that help users manage budgets, savings, and investments effectively (Russell, 2020). AI techniques like natural language processing (NLP) allow financial assistants to understand and respond to user queries conversationally.

2.3 MACHINE LEARNING AND FINANCIAL DATA ANALYSIS

Machine Learning (ML), a subset of AI, uses algorithms to identify patterns and make data-driven predictions. In finance, ML models analyse transaction histories, spending habits, and market trends to provide tailored recommendations and detect anomalies (Hastie, 2009). Supervised learning methods, such as decision trees and neural networks, have proven effective for forecasting user expenses and optimizing saving strategies.

2.4 FINANCIAL LITERACY AND BEHAVIOUR CHANGE

Financial literacy involves understanding financial concepts and applying them to personal money management. Intelligent systems aim not only to provide data but also to improve users' financial literacy by delivering actionable insights and educational content. Behaviour change theories support the design of such systems by emphasizing motivation, habit formation, and feedback loops (Thaler, 2008).

2.5 USER-CENTRED DESIGN AND ACCESSIBILITY

The success of AI financial tools depends on user-friendly interfaces and accessibility, particularly for low-income earners who may have limited experience with technology. Designing for inclusivity requires consideration of simplicity, language support, and mobile accessibility (Norman, 2013). The integration of chatbot interfaces enhances engagement and usability.

2.6 THEORETICAL PERSPECTIVES

This study is grounded in two key theoretical frameworks that explain behavior change through technology: the **Technology Acceptance Model (TAM)** and the **Behavioral Change Theory**.

2.6.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model explains how perceived usefulness and ease of use influence users' acceptance of new technology (Davis, 1989). This theory helps understand how low-income earners in Rwanda might adopt the AI-powered chatbot for financial assistance.

2.6.2 Behavioural Change Theory

The COM-B model, which emphasizes Capability, Opportunity, and Motivation as drivers of Behaviour, guides the design of personalized alerts and reminders to encourage better saving and budgeting habits (Michie, 2011). This theory supports the study's focus on how AI can influence financial behaviour change.

2.6.3 Integration of TAM and Behavioral Change Theory

Combining TAM and COM-B provides a comprehensive framework for examining both technology adoption and behavioural outcomes, enabling this study to analyse how AI chatbots can be effectively used to improve budgeting accuracy, saving discipline, and emergency readiness among low-income earners.

2.7 RELATED STUDIES

Several empirical studies have explored the integration of artificial intelligence in personal finance tools. For instance, a study by (Cao, 2022) examined a financial planning chatbot using natural language processing to offer budgeting advice. The tool increased financial awareness and goal tracking among users, though its effectiveness was limited to individuals with prior digital banking access. Similarly, (Zhang H. &, 2022) developed a mobile application that combined predictive analytics with conversational interfaces, leading to increased user engagement. However, the app targeted urban populations in China, relying heavily on structured banking data.

Other research has highlighted the role of AI in enhancing financial behaviour. (Gomez, 2020) integrated educational content within an AI-based savings assistant, demonstrating improved financial literacy among young adults in Latin America. (Kumar R. S., 2021) evaluated machine learning models for anomaly detection in personal finance apps, emphasizing the value of decision tree and neural network ensembles. Despite their contributions, these studies primarily focus on digitally connected environments. They reveal a literature gap concerning AI tools adapted to informal economies—particularly for underserved users in regions like sub-Saharan Africa. The present study addresses this gap by proposing a culturally and contextually adapted financial assistant for low-income Rwandans.

CHAPTER THREE: DATA COLLECTION, PRESENTATION AND ANALYSIS OF RESULTS

3.1 INTRODUCTION

This chapter outlines the research methodology guiding the study, including the research design, population, sampling methods, data collection instruments, procedures, and techniques for data analysis. It also addresses ethical considerations and potential limitations affecting the validity of the findings. The methodology aims to provide a comprehensive understanding of low-income earners' financial behaviours and evaluates the effectiveness of personal financial assistance systems. Additionally, this chapter describes the current financial guidance systems used by the target population, identifying their limitations, before introducing the proposed system designed to better support users' financial management needs.

3.2 RESEARCH DESIGN

The study adopted a Mixed Methods approach with an Explanatory Sequential Design. This design allows for the collection of both quantitative and qualitative data to provide a comprehensive analysis of the research problem. Initially, quantitative data were gathered through structured surveys to capture patterns in saving and budgeting behaviour. This was followed by qualitative interviews aimed at understanding the motivations, perceptions, and challenges related to personal financial management. The choice of this approach was justified by the need to triangulate findings and enhance the depth of the analysis. The development of FinSmart AI followed the Agile Software Development Lifecycle (SDLC), which involved iterative sprints based on continuous user feedback.(see the Figure 1)

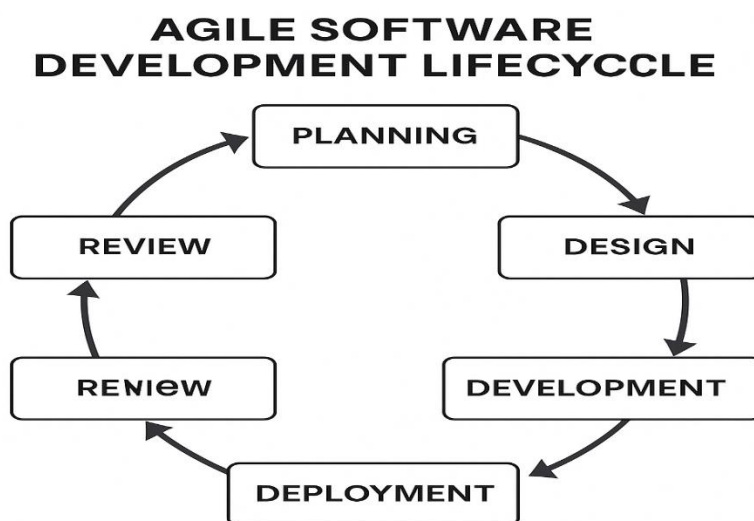


Figure 1: Agile Process

3.3 RESEARCH POPULATION

This study focused on individuals residing in Huye District, particularly those underserved by formal financial institutions. The target population included informal sector workers, university students, and small-scale entrepreneurs—groups that are often vulnerable to financial instability and have limited access to structured financial guidance. The accessible population consisted of digitally literate individuals aged 18 to 45, as they are more likely to engage with digital financial tools.

Based on demographic data, the total population of Huye District is approximately 350,000. Among these, an estimated 40,000 individuals meet the criteria of being aged 18 to 45, digitally literate, and belonging to the target groups outlined above. Participants were selected based on their interest in improving financial habits and their lack of prior use of formal financial advisory platforms. To ensure accurate assessment of the system's impact, individuals already receiving structured financial advice were excluded from the study.

3.4 SAMPLE SIZE AND SAMPLING PROCEDURE

Using the (Krejcie, 1970) method, the ideal sample size for an accessible population of 40,000 is approximately **381 respondents** to achieve a 95% confidence level with a 5% margin of error. To strengthen the validity of the study, a target sample size of **500 respondents** was selected.

Purposive sampling, a non-probability technique, was used to select participants with specific characteristics aligned to the study objectives, including digital literacy and financial vulnerability. This sample size balances statistical rigor with practical considerations, ensuring meaningful insights into the financial behaviors of the target population.

The key parameters used in the sample size calculation are summarized on Table 1:

Parameter	Value
Population size (N)	40,000
Confidence level	95%
Margin of error (d)	5% (0.05)
Z-score (Z^2)	3.841
Population proportion (P)	0.5
Calculated sample size (S)	≈ 381

Table 1: Sampling table

3.5 RESEARCH INSTRUMENT

3.5.1 CHOICE OF THE RESEARCH INSTRUMENT

Two key instruments were employed to collect both quantitative and qualitative data relevant to the research objectives:

Structured Questionnaire: This was used to gather quantitative data on user income, budgeting habits, savings behaviour, and engagement with digital financial tools. The questionnaire items were derived from the variables identified in the literature review and designed to assess the financial behaviour of the target population.

Interview Guide: Semi-structured interviews were conducted to capture in-depth insights into users' experiences, challenges, and attitudes toward AI-powered financial assistance tools. This helped complement and explain trends observed in the quantitative data.

To ensure clarity and consistency, the questionnaire was pre-tested with 20 individuals who were not part of the final sample. Feedback from the pre-test informed revisions to improve item clarity and ensure alignment with the study's objectives

3.5.2 VALIDITY AND RELIABILITY OF THE INSTRUMENT

The validity of the questionnaire was established through expert review by academic supervisors and financial professionals, ensuring relevance and appropriateness of content. Reliability was assessed through the test-retest method, achieving a Cronbach's alpha score of 0.81, indicating high internal consistency.

3.6 DATA GATHERING PROCEDURES

Data collection was conducted in three phases:

Pre-data collection: This involved the development of the research instrument, pre-testing with a pilot group to ensure clarity and reliability, and obtaining ethical approval from the relevant committee.

During data collection: Surveys were administered both online and in physical form to reach a broader respondent base. Interviews were conducted face-to-face and via Zoom to accommodate participants' availability.

Post-data collection: Collected responses were compiled, systematically coded, and organized in preparation for data analysis.

3.7 DATA ANALYSIS AND INTERPRETATION

This study followed an Explanatory Sequential Mixed Methods design, starting with quantitative analysis, followed by qualitative insights to clarify the findings.

Quantitative data from 500 participants were analysed using SPSS. Descriptive statistics (means, frequencies, and percentages) summarized key demographics and financial behaviours, while a Chi-square test examined relationships—such as between income level and saving habits.

Qualitative data from semi-structured interviews were analysed thematically to explore participants' perceptions and experiences with AI-powered financial tools. Together, the results offered a comprehensive view of FinSmart AI's effectiveness in promoting trust, usability, and financial behaviour change. (see Table 2)

Data Type	Variables Analysed	Analysis Technique	Key Findings (Sample)
Quantitative	Age, income, budgeting habits	Descriptive statistics	Mean age = 28 years; 60% save regularly
Quantitative	Income level & saving behaviour	Chi-square test	$\chi^2 = 8.45$, $p < 0.05$ (significant relationship)
Qualitative	Interview transcripts	Thematic analysis	Themes: trust in AI, ease of use, financial awareness

Table 2: Data Analysis

The average age of respondents was 28 years, with 60% reporting regular saving habits. A statistically significant relationship was found between income level and saving behaviour ($\chi^2 = 8.45$, $p < 0.05$), indicating that income influences saving practices among participants. Thematic analysis of interviews highlighted trust in AI, ease of use, and increased financial awareness as major themes impacting users' acceptance of the FinSmart AI tool.

3.8 ETHICAL CONSIDERATIONS

All participants provided informed consent prior to the study. Personal data were kept confidential and used only for research purposes. Participation was voluntary, and individuals could withdraw at any time without consequences. The study ensured that the safety, privacy, and well-being of all participants were protected throughout the research process.

3.9 LIMITATIONS OF THE STUDY

This study has several limitations that may affect the validity of its findings:

- ❖ Purposive sampling may introduce bias and limit how well results represent the wider population.
- ❖ Limited financial data, varied digital literacy, and unreliable internet access—especially in rural areas—may have restricted participation.
- ❖ The FinSmart AI system is currently web-based and does not support Android or mobile platforms, which may limit accessibility and user engagement.

Despite these limitations, multiple data collection methods and careful instrument design helped improve data quality and reduce bias. Acknowledging these issues informs readers about the trustworthiness of the findings

3.10 DESCRIPTION OF THE CURRENT SYSTEM

Many low-income earners in Rwanda rely on manual expense tracking methods that are often inaccurate and inconsistent, making budgeting and financial planning. This reliance leads to poor financial decisions, irregular savings, and increased vulnerability. Current financial tools lack personalized advice and support for behavioural change, limiting their effectiveness.

3.11 PROPOSED SYSTEM

FinSmart AI is an intelligent personal financial assistant that uses AI to analyze users' spending behaviour and provide personalized advice. The system tracks income and expenses, evaluates financial habits, and sends tailored notifications to encourage better spending decisions. Accessible via web platforms, FinSmart AI aims to improve financial awareness and promote inclusive growth among underserved communities in Rwanda.

System Architecture Overview

The system is composed of a web-based front end (built using HTML, CSS, PHP), a Flask-powered AI chatbot back end, and a MySQL/PostgreSQL database. An integrated notification engine sends email alerts. Data is collected through structured forms and AI-led conversations, then stored securely. NLP and Open AI is used to interpret user queries and provide smart financial guidance.

3.11.1 Use Case Diagram

The Use Case Diagram illustrates the interactions between different user roles and the FinSmart AI system. Key actors include the regular user and the administrator. Users can perform core functions such as setting a budget, chatting with the AI assistant, and receiving personalized financial tips. The admin oversees the system, manages content, and ensures user support look on Figure 2.

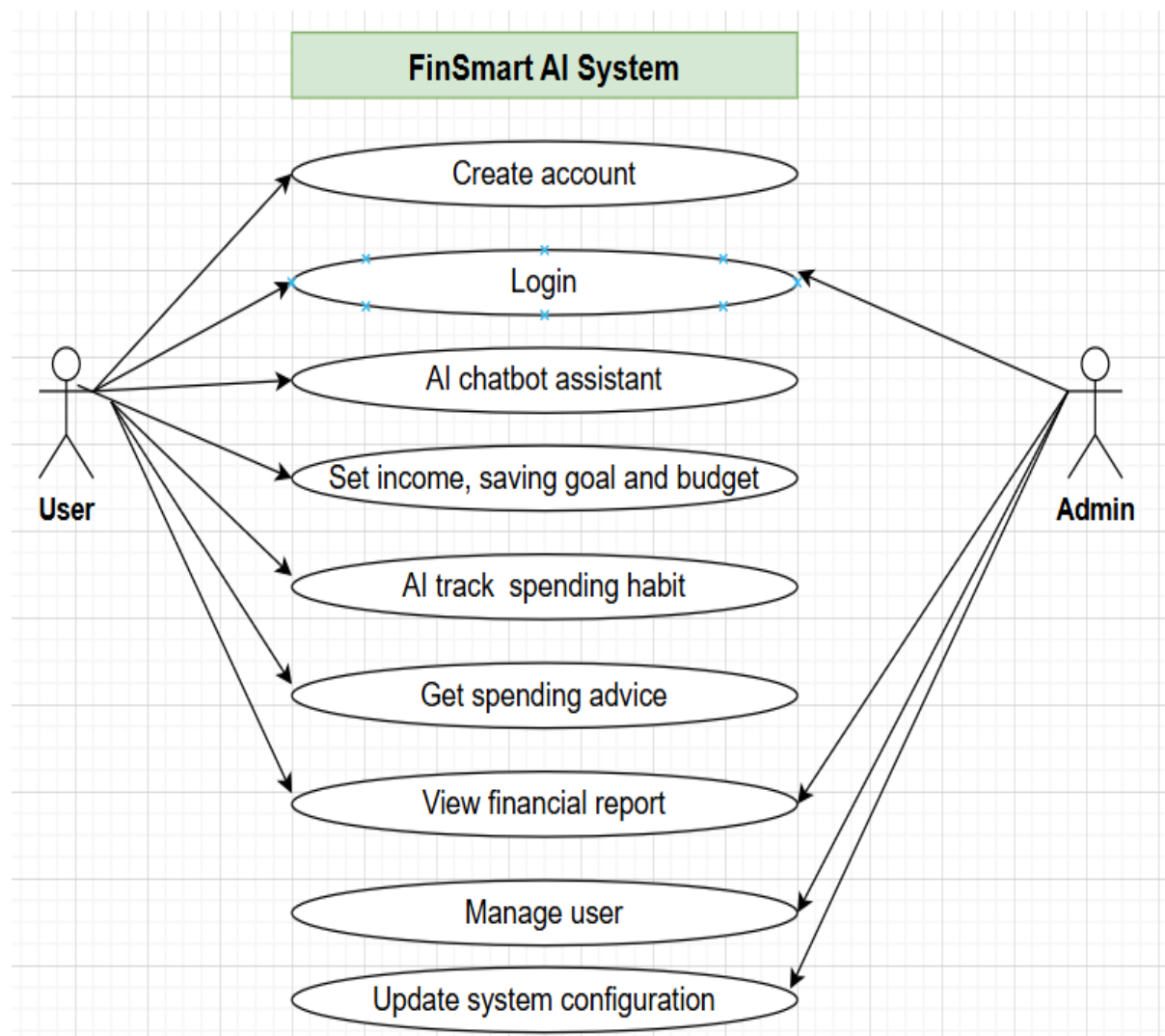


Figure 2(Use case Finsmart AI)

3.11.2 Context Diagram

The context diagram shows how external users interact with the FinSmart AI system. The user manually enters financial data such as income, budget expenses, and savings goals through a chatbot. In return, the system provides spending advice, transaction report, and email alerts. The administrator manages the system by updating settings and maintaining the overall functionality to ensure accurate and helpful financial guidance(see Figure 3).

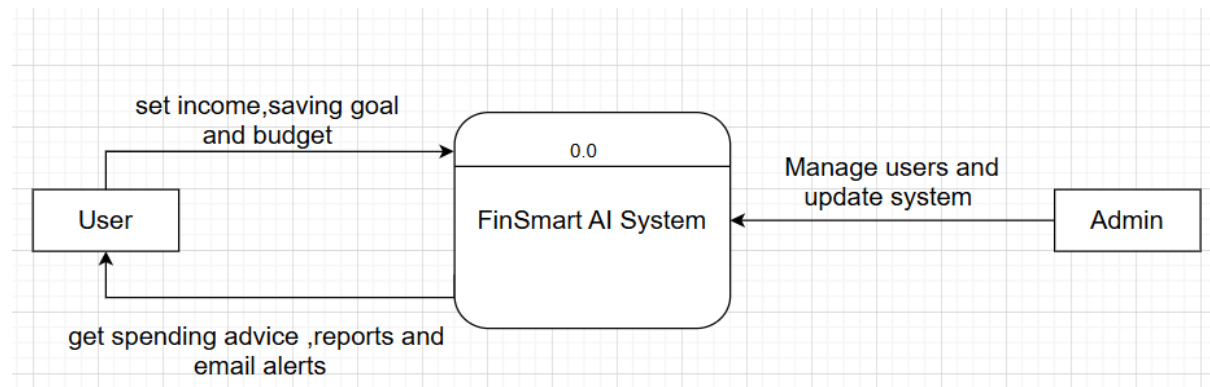


Figure 3: Context Diagram

3.11.3 Data Flow Diagram

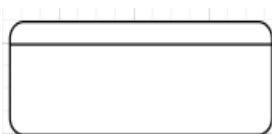
A Data Flow Diagram (DFD) graphically represents the flow of data within the FinSmart AI system. It illustrates how users interact with the system, how processes transform data, where data is stored, and how outputs such as reports and notifications are generated. The DFD shows **entities, processes, data stores, and data flows**, providing a clear overview of the system's logic and operations .(look on Figure 4)

Key Concepts for This DFD:

- **Entities** : Users and Admin who interact with the system



- **Processes** : Tasks like registration, budgeting, spending tracking, analysis, and recommendations



- **Data Stores** –:Storage of user info, financial data, budgets, transactions, and email logs



- **Data Flows** – Movement of data between entities, processes, and storage



- **Analysis & Reports** – Budget Analysis Engine and Chatbot provide insights
- **Notifications** – System-generated alerts and emails

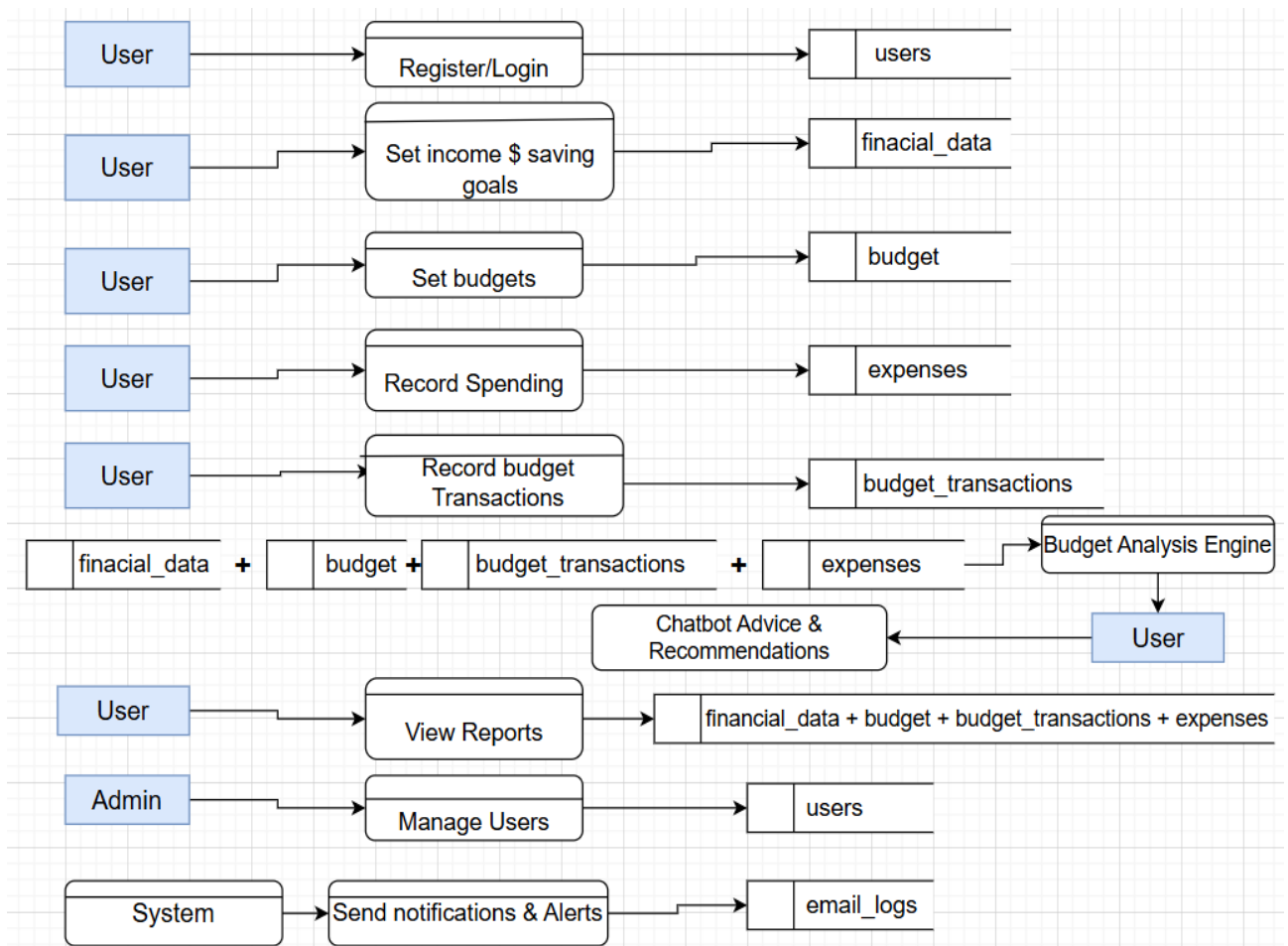


Figure 4: Data Flow Diagram(Finsmart AI)

3.11.4 ERD – Entity Relationship Diagram

The Entity–Relationship Diagram (ERD) illustrates the conceptual structure of the FinSmart AI database. It identifies the main entities—User, Budget, Expenses, FinancialData, BudgetTransactions, ChatbotInteractions, EmailLogs, and TargetSavings—and shows how they are related. The User is the central entity connected to all others, reflecting their role in recording expenses, managing budgets, tracking savings, and interacting with the system. This model ensures logical data flow and supports the implementation of the system’s database.(see Figure 5)

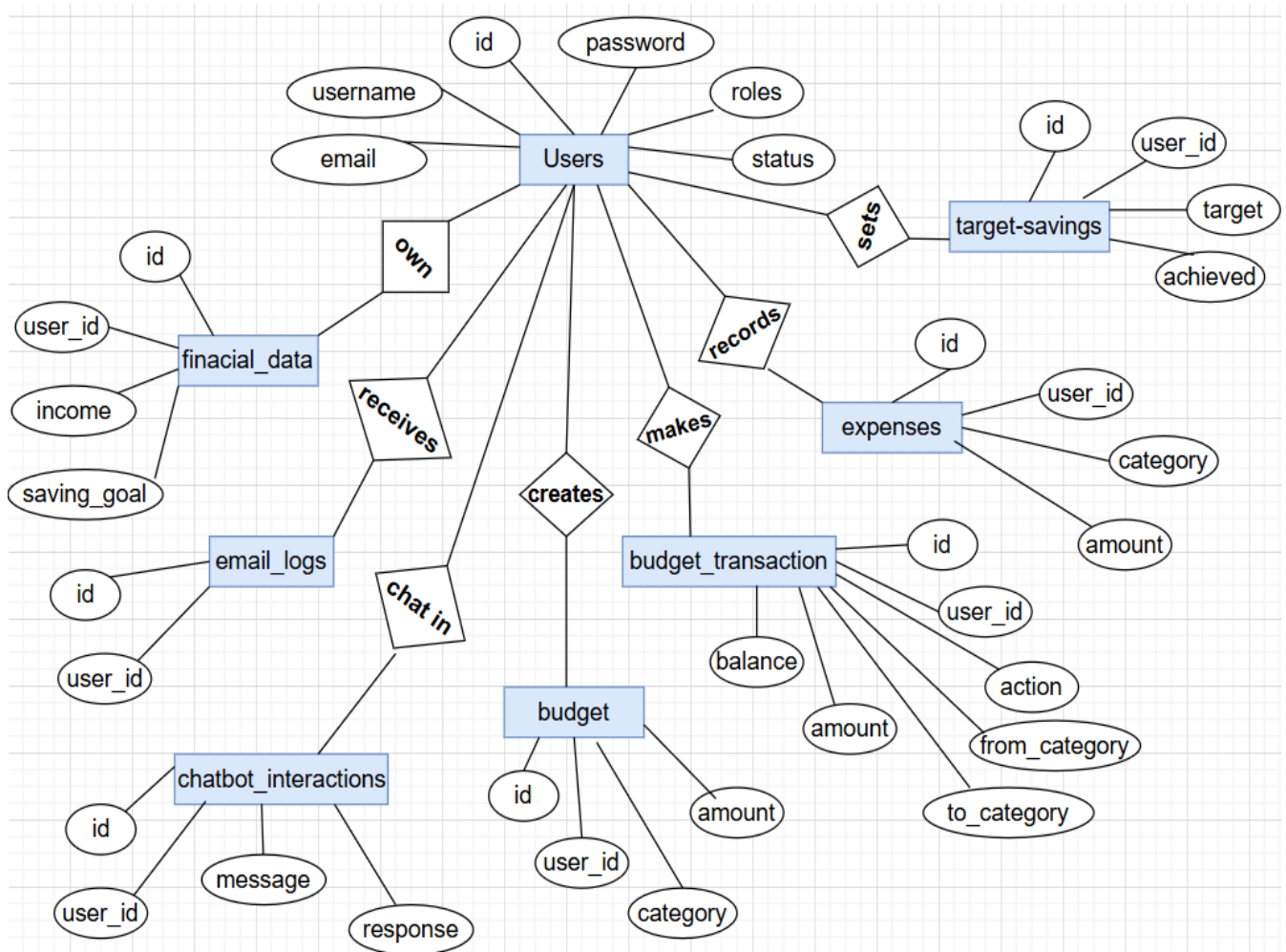


Figure 5: ERD Diagram(finsmart AI)

3.11.5 Physical Data Modeling

The Physical Entity–Relationship Diagram (Physical ERD) represents the actual implementation of FinSmart AI the database structure. Unlike the conceptual model, it defines tables, columns, primary keys, and foreign keys, showing how data is stored and linked in the system. It translates the logical relationships into a design ready for database creation, ensuring consistency, referential integrity, and efficient storage.(see Figure 6)

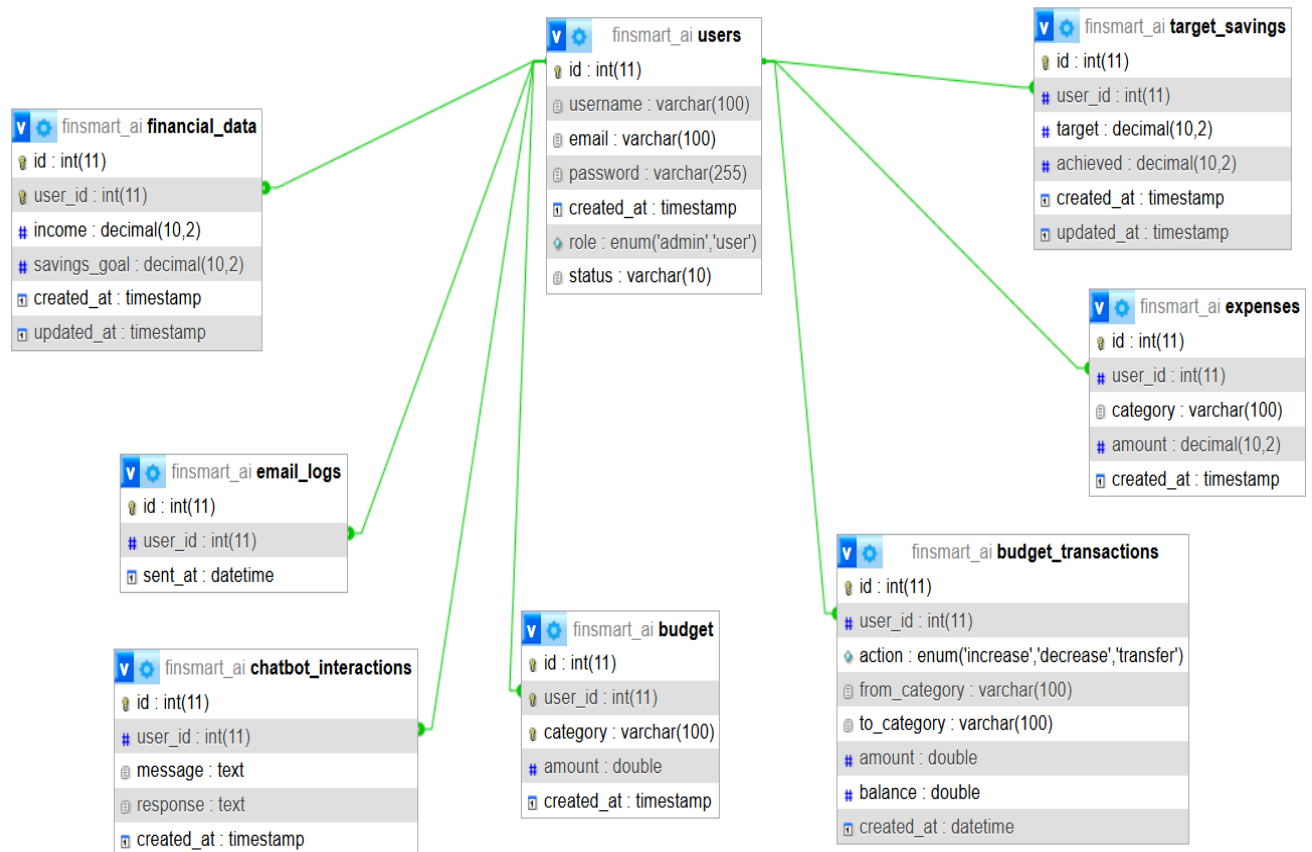


Figure 6: Physical data modeling

3.12 DATA DICTIONARY

A data dictionary is a systematic listing of all data elements relevant to the FinSmart AI system, providing clear and precise definitions. It serves as a common reference for users and system analysts to ensure a shared understanding of all inputs, outputs, and components involved in the financial management chatbot. This organized documentation facilitates accurate data handling, consistent communication, and effective system design throughout the development and deployment of the FinSmart AI tool (Loney, 2009)

Table :user Stores all registered system users, including login credentials and basic profile information. (see Table 3)

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	int(11)	Primary key,No null, Auto_increment	User identification
username	varchar(100)	No	User name
email	varchar(100)	No	User email
password	varchar(255)	No	User password
created_at	timestamp	No	current_timestamp()
role	enum('admin', 'user')	Yes	User role
status	Varchar(10)	No	User status active or inactive

Table 3: user

Table :chatbot_interactions Stores conversations between users and the chatbot, including messages and system-generated responses.(see Table 4)

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	int(11)	Primary key,No null, Auto_increment	Chabot identifier
user_id	int(11)	Yes	User identification(foreign key)
message	text	Yes	User request
response	text	Yes	Chatbot response
created_at	timestamp	No null	current_timestamp()

Table 4:Chatbot

Table :financial_data Keeps records of each user’s income, savings goals, and other financial inputs that form the basis for budgeting and analysis.(see Table 5)

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	int(11)	Primary key,No null, Auto_increment	Financial data identifier
user_id	int(11)	Yes	User identification(foreign key)
income	decimal(10,2)	Yes	User income
savings_goal	decimal(10,2)	Yes	User saving goals
created_at	timestamp	No null	current_timestamp()
updated_at	timestamp	No null	current_timestamp()

Table 5: Financial data

Table: expenses Logs daily spending records, linked to categories, for monitoring and reporting purposes(see Table 6)

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	int(11)	Primary key,No null, Auto_increment	Expenses Identifier
user_id	int(11)	Yes	User identification(foreign key)
category	varchar(100)	Yes	NULL
amount	decimal(10,2)	Yes	NULL
created_at	timestamp	No null	current_timestamp()

Table 6: Expenses

Table: target_savings Holds user-defined saving goals, including target amounts and progress tracking.(see Table 7)

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	int(11)	Primary key,No null, Auto_increment	Target saving Identifier
user_id	int(11)	Yes	User identification(foreign key)
target	decimal(10,2)	Yes	User saving goal target
achieved	decimal(10,2)	Yes	User Saving goal achieved
created_at	timestamp	No null	current_timestamp()
updated_at	timestamp	No null	current_timestamp()

Table 7:target savings

Table: budget Holds the user-defined budget allocations for different categories (e.g., food, transport, savings) observe Table 8

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
id	int(11)	Primary key,No null, Auto_increment	budget Identifier
user_id	int(11)	Yes	User identification(foreign key)
category	Varchar(20)	No null	Budget category
amount	double	No null	Amount of category
created_at	timestamp	No null	current_timestamp()

Table 8: budget

Table: budget_transaction Tracks adjustments or movements within budgets, such as transfers between categories or updates to allocations. (see Table 9)

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
Id	int(11)	Primary key,No null, Auto_increment	budget interaction Identifier
user_id	Int(11)	No null	User identification(foreign key)
Action	Enum(increase,'decrease','transfer')	No null	Budget category
from_category	Varchar(100)	yes	Form which category
to_category	Varchar(100)	yes	To which of category

amount	double	yes	Amount of category
balance	double	yes	Remain balance
created_at	datetime	yes	current_timestamp()

Table 9: budget_transaction

Table: email_logs Maintains a record of notification emails sent to users, including timestamps for tracking communication(see Table 10)

FIELD NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
Id	int(11)	Primary key, No null, Auto_increment	email log Identifier
user_id	int(11)	Yes	User identification(foreign key)
Sent_at	datetime	yes	current-timestamp()

Table 10: email_logs

CHAPTER FOUR: DESIGN AND IMPLEMENTATION

4.1 INTRODUCTION

This chapter outlines the design and implementation process of the FinSmart AI system, a personal finance assistant built to help users manage their income, expenses, savings, and budgeting habits through a conversational chatbot interface. The chapter presents the tools and technologies used, the system architecture, key modules, and the design elements implemented. It also includes screenshots that represent the operational aspects of the system and discusses the outcome and challenges experienced during the development process.

4.2 SYSTEM IMPLEMENTATION OVERVIEW

The implementation of FinSmart AI involved developing a web-based system that integrates natural language processing (NLP), OpenAI-powered conversational logic, a financial tracking engine, and a responsive user interface. The system was developed using PHP for frontend design, with Python and Flask handling the backend processes. A trained chatbot was implemented to interpret finance-related queries and provide smart responses using both predefined intents and OpenAI APIs. Users can securely input and manage their personal financial data through an intuitive interface, ensuring both ease of use and intelligent feedback.

4.3 TECHNOLOGIES AND TOOLS USED

4.3.1 Software Tools

- ❖ **Python Libraries & Flask Framework:** Used to implement the chatbot engine, backend logic, and API endpoints.
- ❖ **NLP (Natural Language Processing):** In FinSmart AI, NLP enables the chatbot to interpret user financial queries using natural language, making interactions smooth and intuitive.
- ❖ **OpenAI API:** Powers FinSmart AI's conversational intelligence by generating accurate and context-aware financial responses, enhancing the user's experience through smart dialogue.
- ❖ **PHP:** Designed the frontend user interface and ensured seamless interaction with backend endpoints.
- ❖ **HTML, CSS, JavaScript:** Created a responsive and dynamic interface, enhancing user experience.
- ❖ **MySQL Database (via XAMPP):** Handles secure storage of users, expenses, budgets, and budget_transactions, emails logs, chatbot interactions and financial data.

- ❖ **Lucidchart:** Used for use case diagrams and sequence diagram representations.
- ❖ **Jupyter Notebook:** Used for building and testing FinSmart AI's machine learning model like Decision Tree, Random Forest for financial behavior prediction.

4.3.2 Hardware Tools

- ❖ Standard development laptops ($\geq 8\text{GB}$ RAM) with stable internet for testing and deployment.
- ❖ The system was tested desktop browsers to ensure full cross-platform support.

4.4 SYSTEM ARCHITECTURE AND KEY COMPONENTS

- ❖ **Frontend Layer:** Built using PHP, HTML, CSS, and JavaScript, it provides user input forms and the chatbot interface.
- ❖ **Backend/API Layer:** Built in Python, this layer handles core system logic, machine learning-based prediction, NLP processing, and financial computations. Flask is used during integration with the frontend for serving API endpoints.
- ❖ **Database Layer:** Manages structured user data in MySQL, supporting persistence and query operations.
- ❖ **OpenAI/NLP Layer:** Handles interpretation of user intents and generates intelligent natural language responses.

4.5 CORE FUNCTIONAL MODULES

- ❖ **Chatbot Module:** Enables two-way communication, interpreting user financial queries and offering intelligent replies.
- ❖ **Budget Tracker:** Provides visual and numeric summaries of user budgets and spending patterns.
- ❖ **Income & Expense Logger:** Facilitates the recording of transactions with tagging by category, time, and purpose.
- ❖ **Financial Advice Engine:** Analyzes user habits and suggests strategies for emergency and financial planning.

4.6 DATASET DESCRIPTION

The dataset contains 93 anonymized financial records with features such as income, expenses, and linguistic indicators. To address class imbalance between users who met and did not meet their goals, oversampling was applied to balance the classes before training.

4.6.1 Training Configuration

Developed in Jupyter Notebook (Python 3.10), the system uses Pandas and NumPy for data handling, Scikit-learn and Imbalanced-learn for preprocessing and modeling, spaCy for NLP feature extraction, Matplotlib and Seaborn for visualization, GridSearchCV for hyperparameter tuning, and Joblib for saving models.

4.6.2 Model Training ,Selection,Testing And Performance Evaluation

Two classification models—Decision Tree and Random Forest—were trained and tuned using GridSearchCV. The Random Forest classifier achieved the highest test accuracy of 100%.(see Figure 7: Balanced Class

Figure 8:

Confusion Matrix)

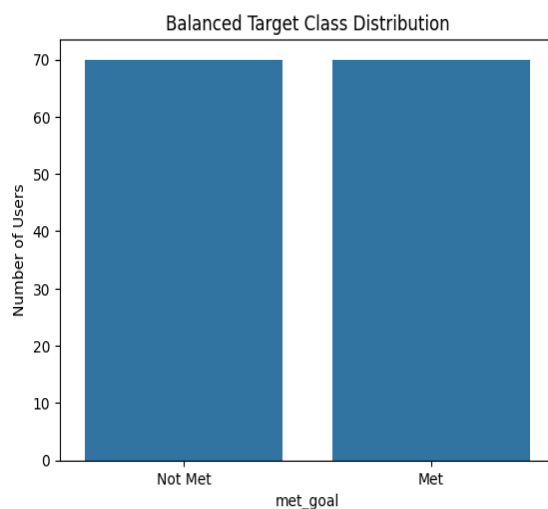


Figure 7: Balanced Class

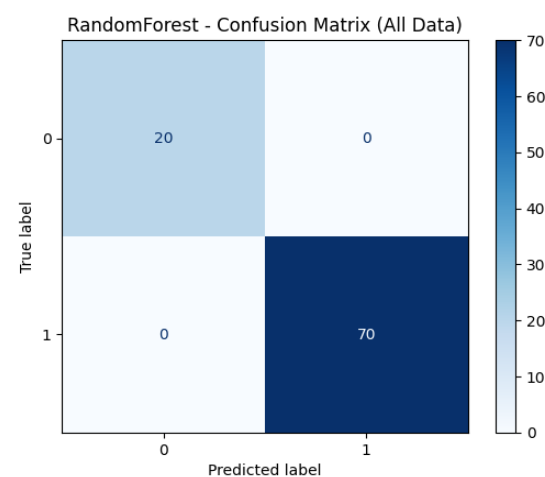


Figure 8: Confusion Matrix

Random Forest model achieved a training accuracy of 1.00 and a testing accuracy of 1.00, showing consistent and reliable classification performance.(see Figure 9)

```
RandomForest Training Accuracy: 1.0000
RandomForest Test Accuracy: 1.0000
```

Figure 9:Random forest

Decision Tree model reached a training accuracy of 0.9861 and a testing accuracy of 1.00, demonstrating high accuracy and good generalization to unseen data(see Figure 10)

```
DecisionTree Training Accuracy: 0.9861
DecisionTree Test Accuracy: 1.0000
```

Figure 10:Decision tree

4.7 SCREENSHOTS OF THE SYSTEM

4.7.1 Welcome Page

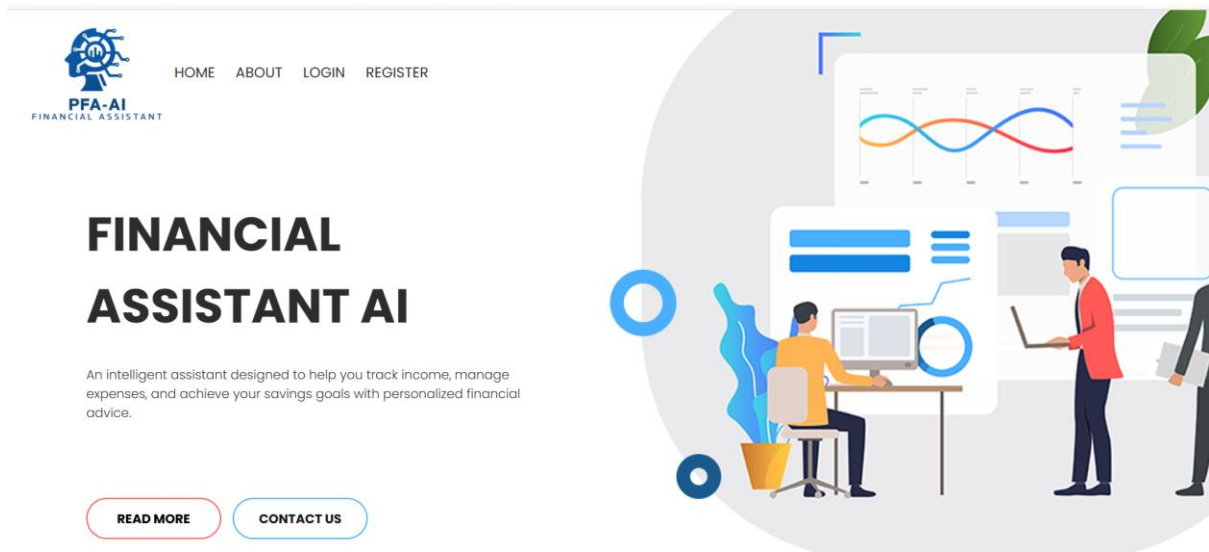


Figure 11: Welcome Page

4.7.2 User Registration and Login Interface

Secure forms allowing new users to register and existing users to log in using username/email and password.(see Figure 12: user registration

Figure 13: user

login)

Create an Account

Username:

Email:

Password:

[Register](#)

[back](#) Already have an account? [Login here](#)

Figure 12: user registration

Login to Your Account

Email:

Password:

[Login](#)

[Back](#) | Don't have an account? [Register here](#)

Figure 13: user login

4.7.3 Dashboard – Summary of User Spending

On first login, the dashboard shows zero for all sections since no data has been added yet (empty, initial, overview). (see Figure 14)

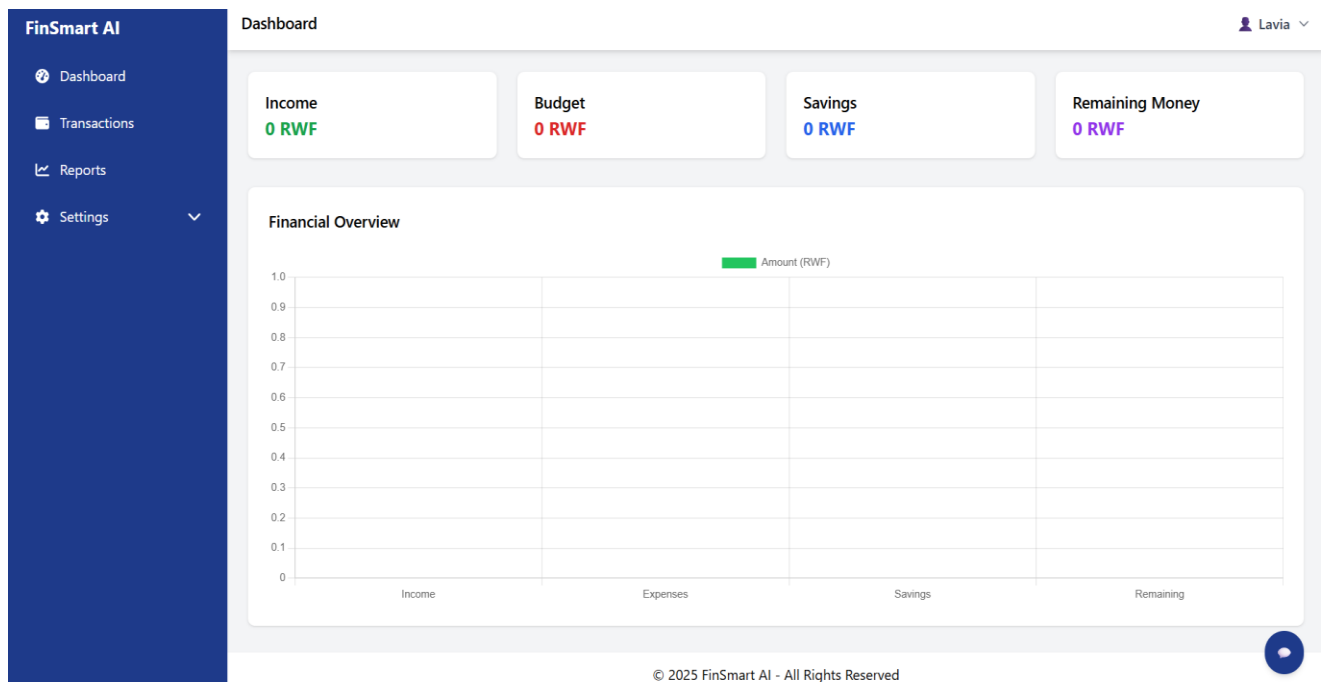


Figure 14: dashboard 1

After updating their income and setting budget and saving goals, Dashboard updates to show the user's income, budget, savings, and remaining money. (current, updated, status) Figure 15

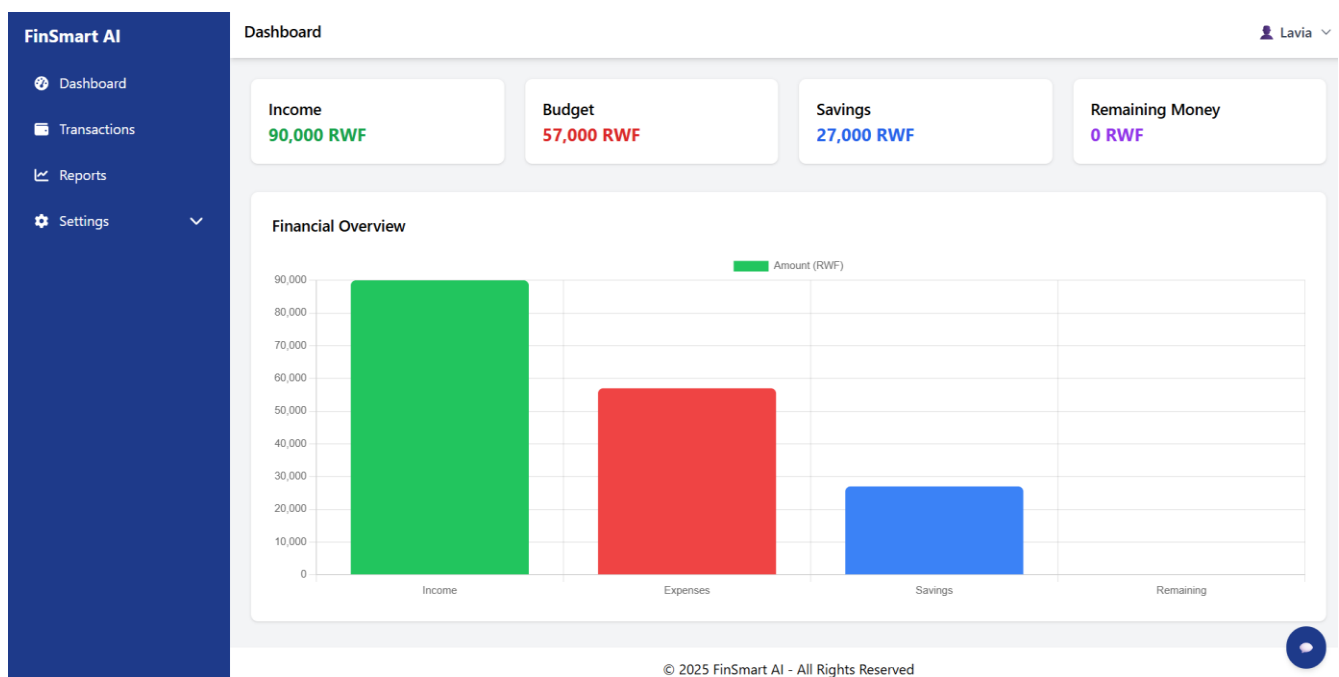


Figure 15: dashboard 2

4.7.4 Chatbot Interface

The user interacts with the Finsmart AI chatbot to get help managing their finances. The system asks the user to provide their current monthly income, saving goals, and budget (interactive, guidance, input). See Figure 16

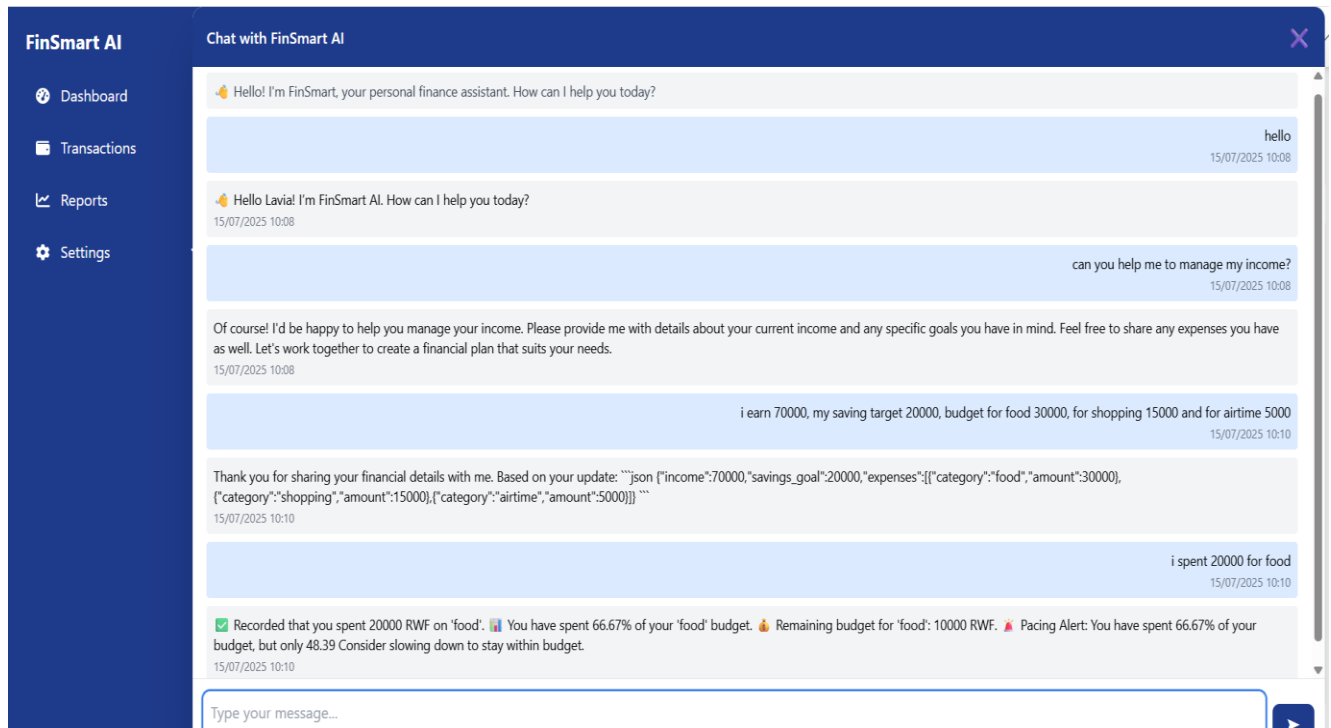


Figure 16: Chatbot interaction 1

When the user has a financial emergency, FinSmart AI suggests which top spending category to reduce, displays the updated balance, and enables transferring amounts between budget categories to manage funds efficiently. See Figure 17

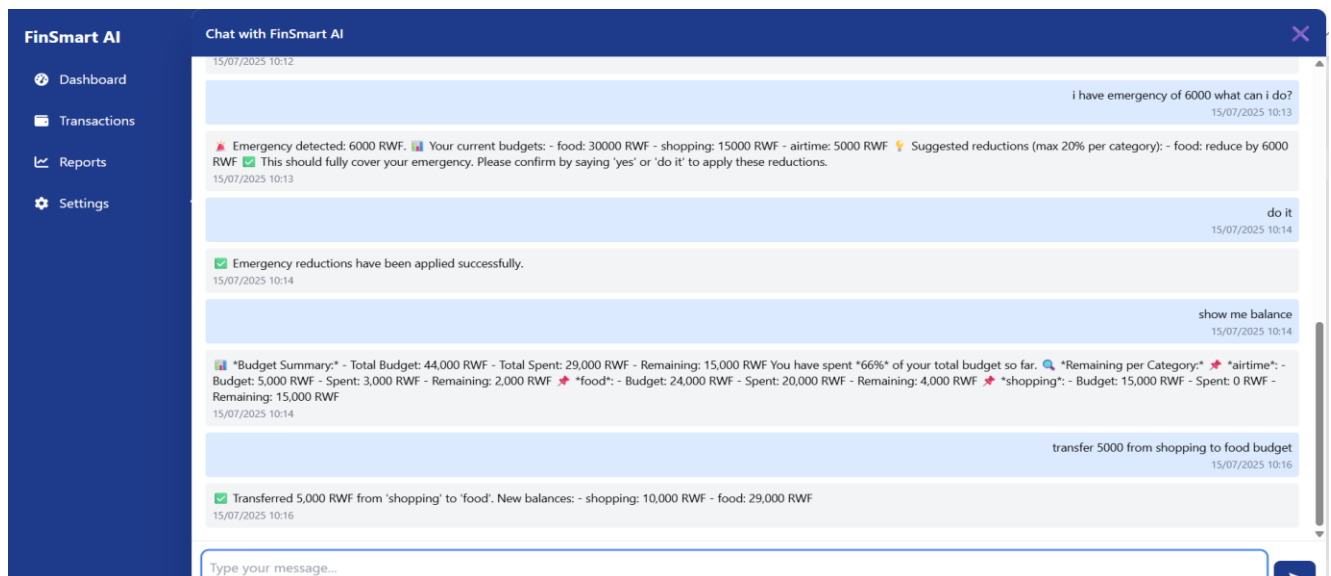


Figure 17: Chatbot interaction 2

The user spends money in different budget categories(shopping,airtime), updates income, and increases budgets for existing categories (food). See Figure 18



Figure 18: chatbot interaction 3

The user increases their saving goals and adds a new budget for haircut, updating FinSmart AI to reflect the changes (savings, new budget, update). (see Figure 19)

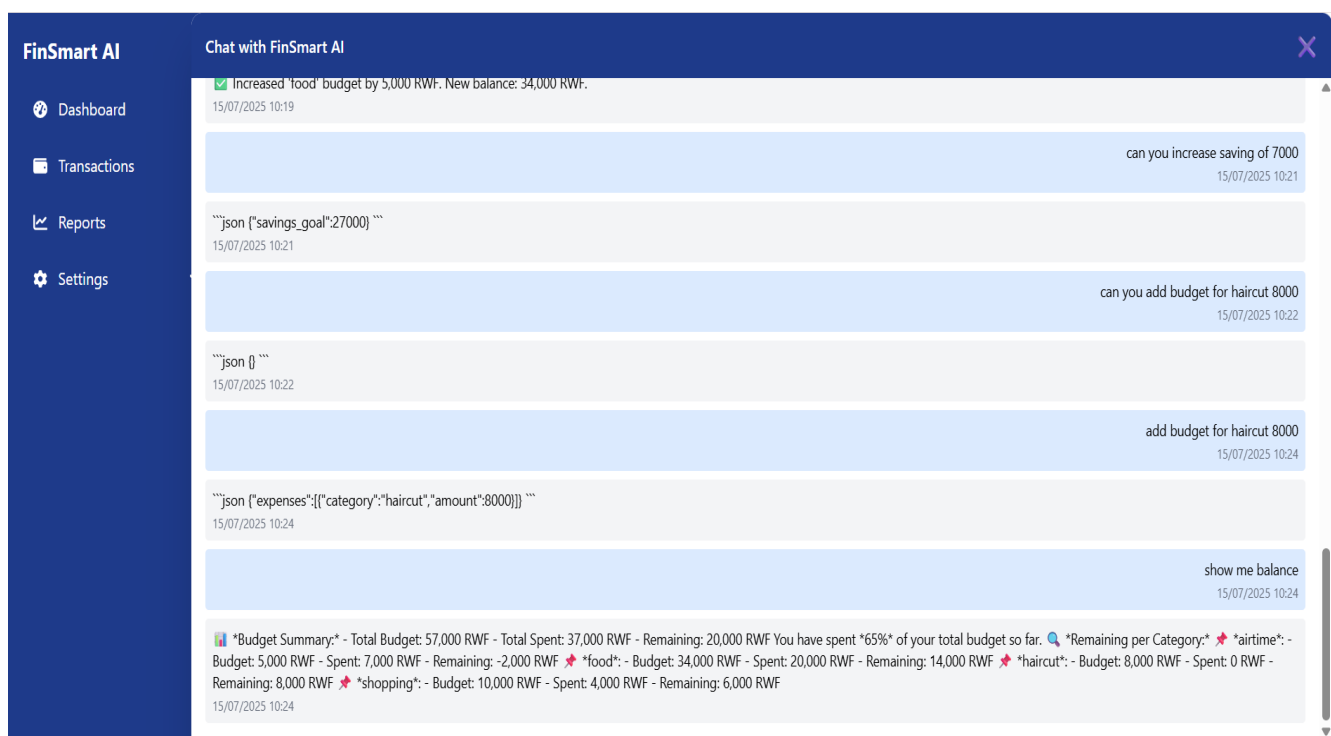


Figure 19: chatbot interaction 4

4.7.5 Transaction Section

On first login, the transaction table is empty since no activity has been recorded yet (empty, initial, overview).(see Figure 20)

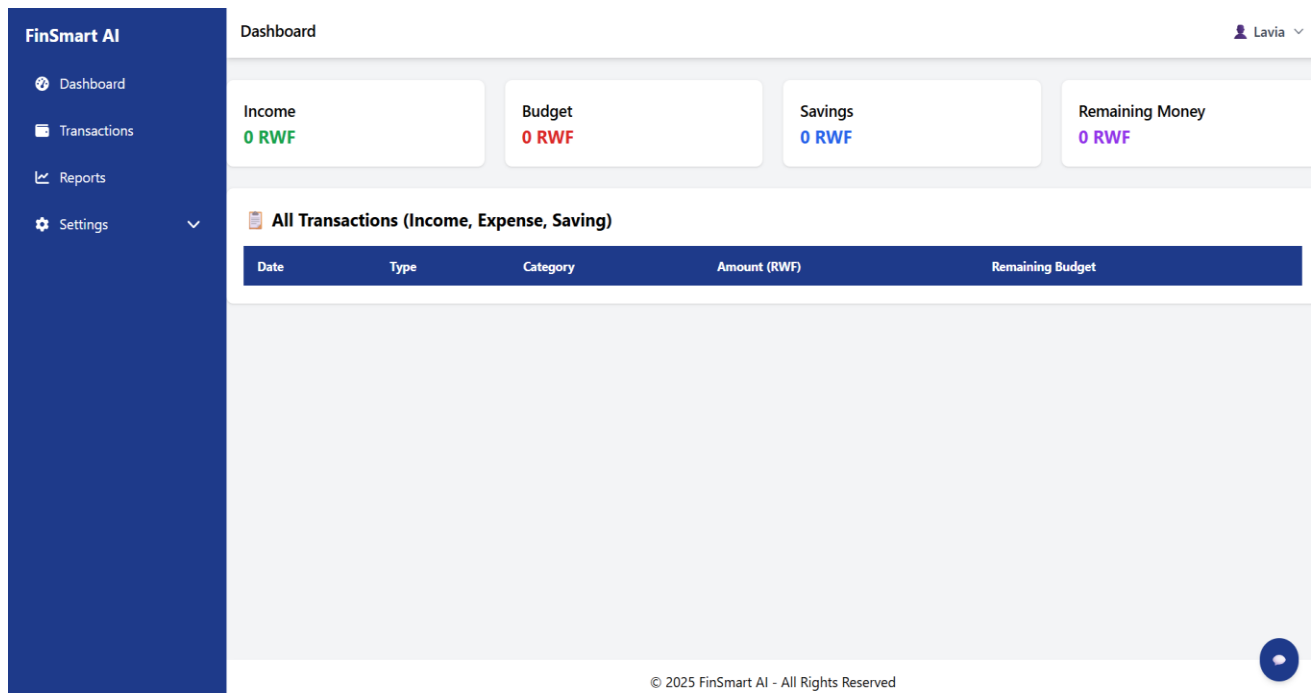


Figure 20:Transaction 1

After setting income, savings, and recording spending, every transaction is captured and displayed in the transaction table (recorded, tracked, updated).(see Figure 21)

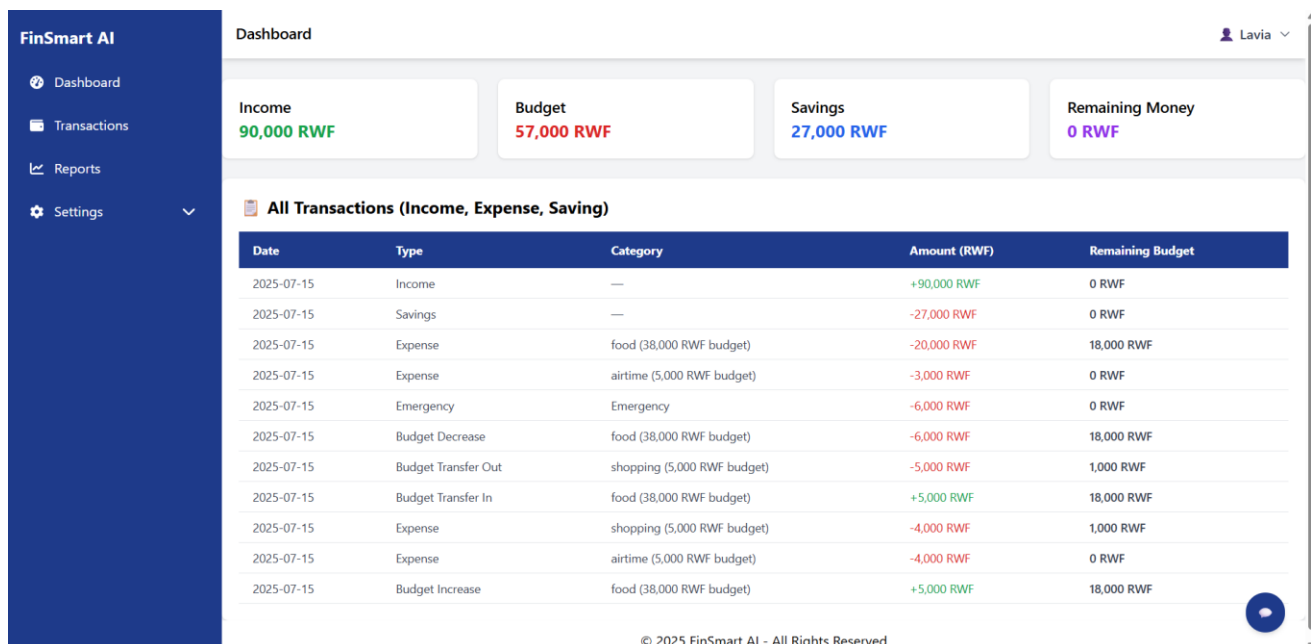


Figure 21: Transaction history

4.7.6 Report Section

On first login, the report section is empty since no financial activity has been recorded yet (empty, initial, overview).(see Figure 22)

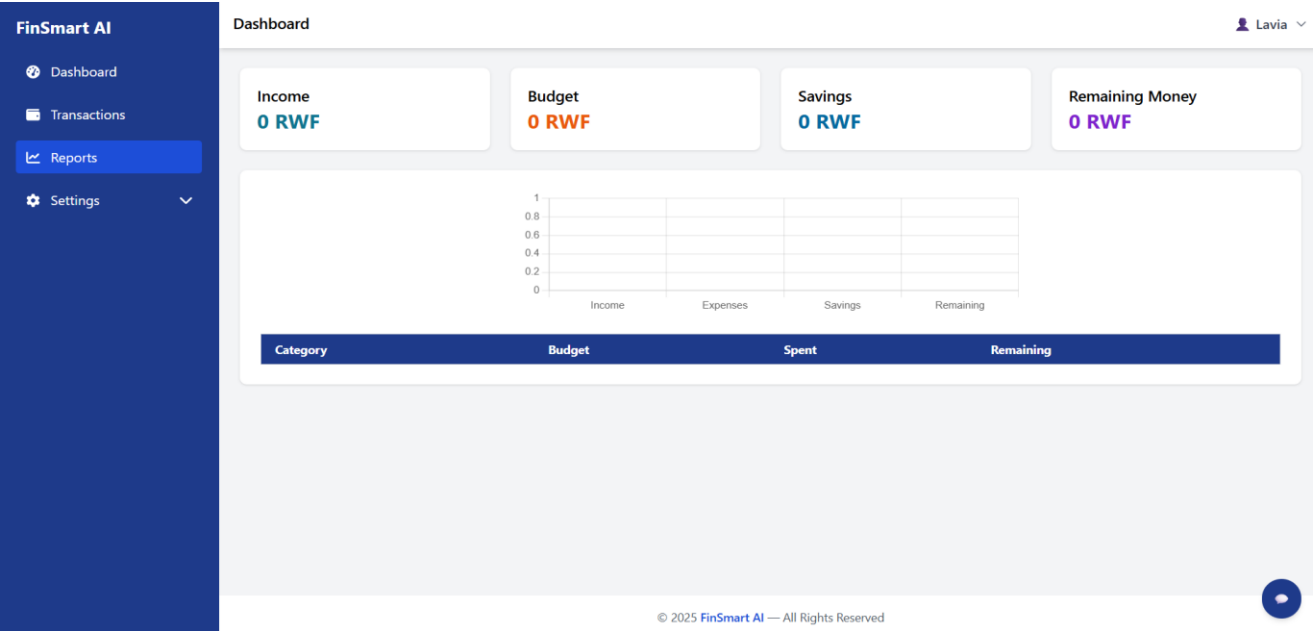


Figure 22: Report1

After updating income, savings, and recording spending, the system generates reports showing all financial activities and current status (generated, tracked, updated).(see Figure 23)

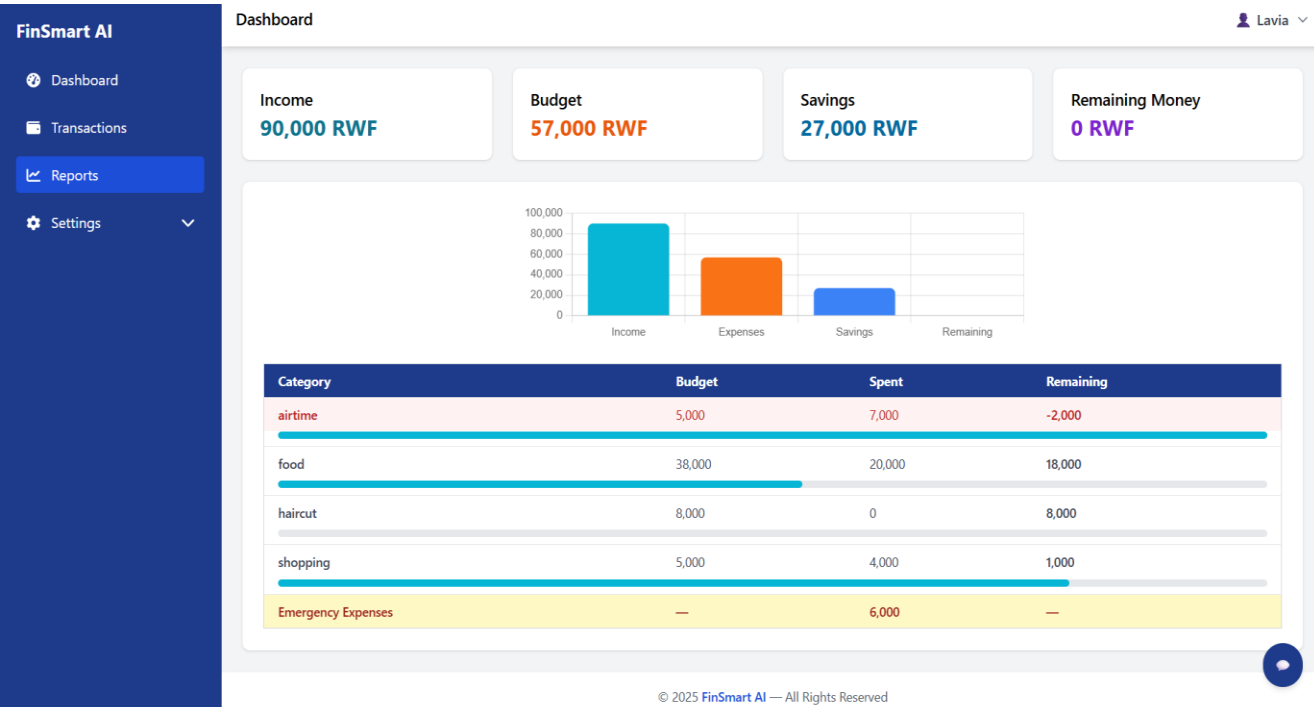


Figure 23: Report 2

4.7.7 Email Notification Section

Users receive important email notifications related to their financial activity, alerts, and system messages generated by FinSmart AI on Figure 24

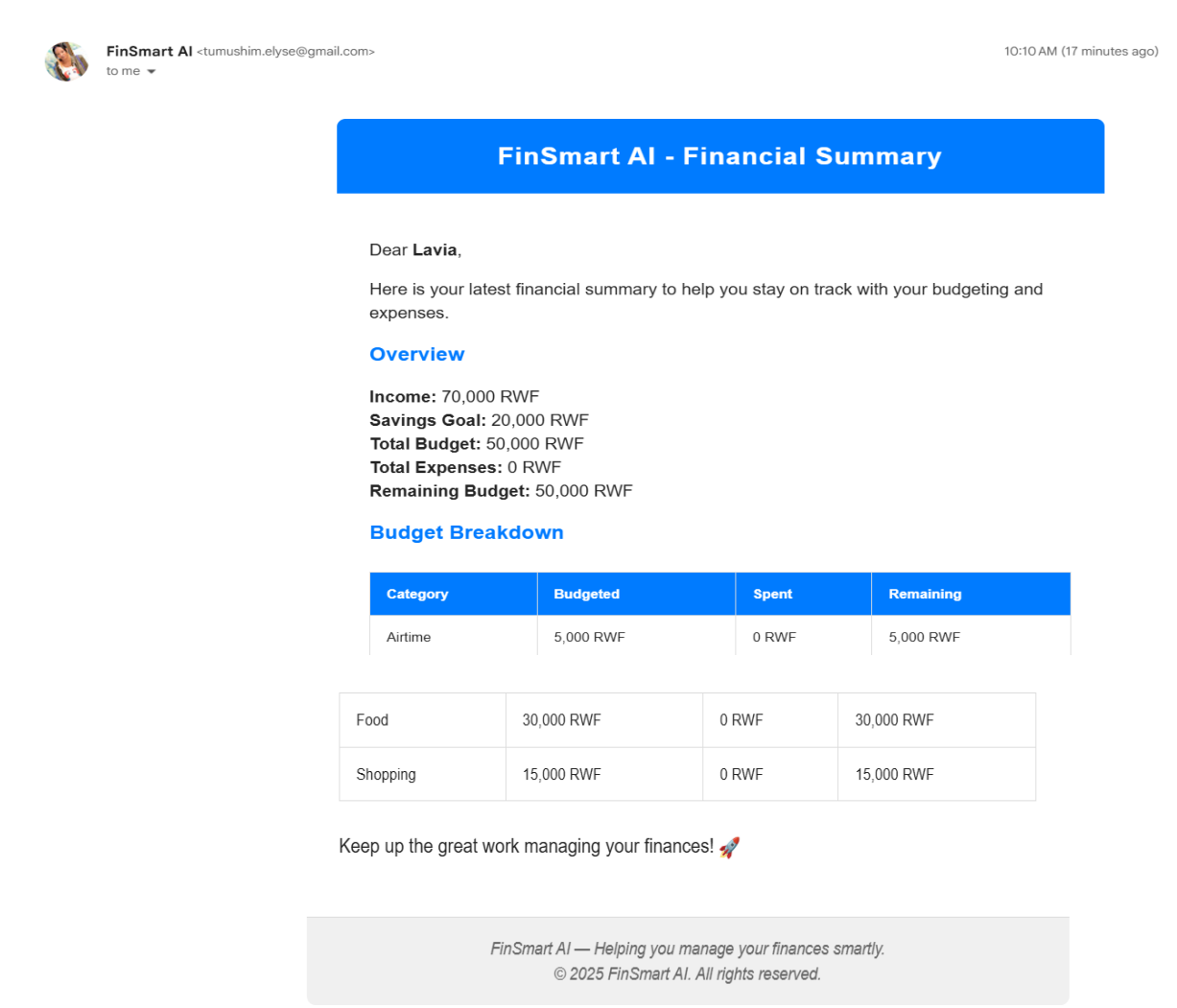


Figure 24:Email Notification Before

After spending, income updates, and budget changes, the user receives an email summarizing all activities, showing updated financial information (notification, summary, activity) (see Figure 25)

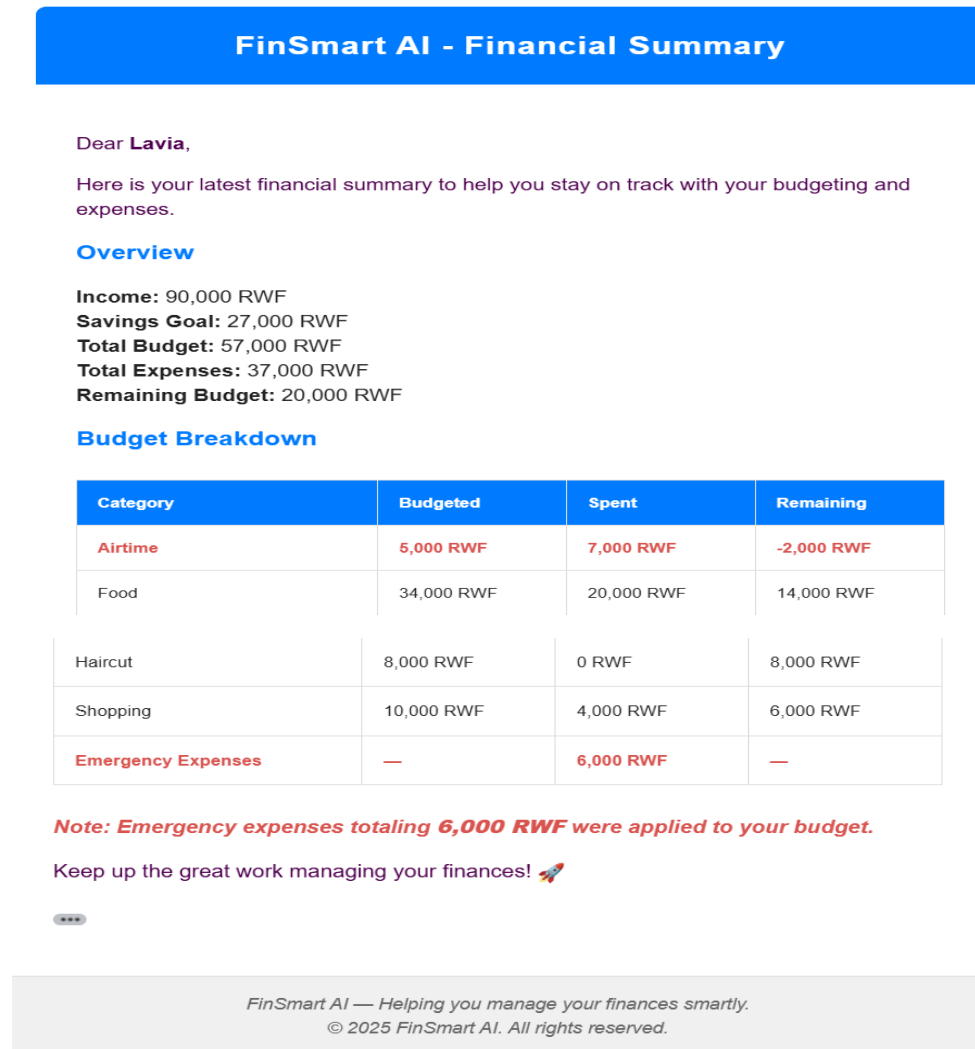


Figure 25:Email Notification after

When the user logs back into the system after some time and makes changes such as updating income, spending, or budgets, they receive an email summarizing their recent activities. The email shows each transaction with date and time, highlights top spending categories with recommendations, and includes a downloadable PDF report containing all this information (recent, summary, downloadable) view Figure 26

FinSmart AI - Financial Summary

Dear **Lavia**,

🎉 You are managing your finances well!

Overview

Income: FRW 90,000.0
Savings Goal: FRW 27,000.0
Total Spent: FRW 31,000.0
Top Spending Category: food
Tip: 🛒 Buy staples in bulk to save money.

Budget vs Spending

Category	Budgeted	Spent	Remaining
Airtime	5,000.0 RWF	7,000.0 RWF	-2,000.0 RWF
Food	34,000.0 RWF	20,000.0 RWF	14,000.0 RWF
Haircut	8,000.0 RWF	0.0 RWF	8,000.0 RWF
Shopping	10,000.0 RWF	4,000.0 RWF	6,000.0 RWF

Recent Expenses

Category	Amount	Date
Airtime	4,000.0 RWF	2025-07-15 10:17:36
Shopping	4,000.0 RWF	2025-07-15 10:17:13
Emergency	6,000.0 RWF	2025-07-15 10:14:05
Airtime	3,000.0 RWF	2025-07-15 10:12:08
Food	20,000.0 RWF	2025-07-15 10:10:51

FinSmart AI — Helping you manage your finances smartly.
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One attachment • Scanned by Gmail ⓘ

User Login
 Income FRW 90000.0
 Total Spent FRW 31000.0
 Savings Goal FRW 27000.0
 Top Spending Category: food
 Tip: Buy staples in bulk to save money.

Recent
 Summary
 Downloadable

financial_report_...

Figure 26: Email for recently user login

4.7.8 Admin Section

The admin can view all users' total income, expenses, and savings, as well as the complete list of users. The dashboard allows managing accounts, including activating or deactivating users and monitoring overall financial activity (management, overview, control). View Figure 27

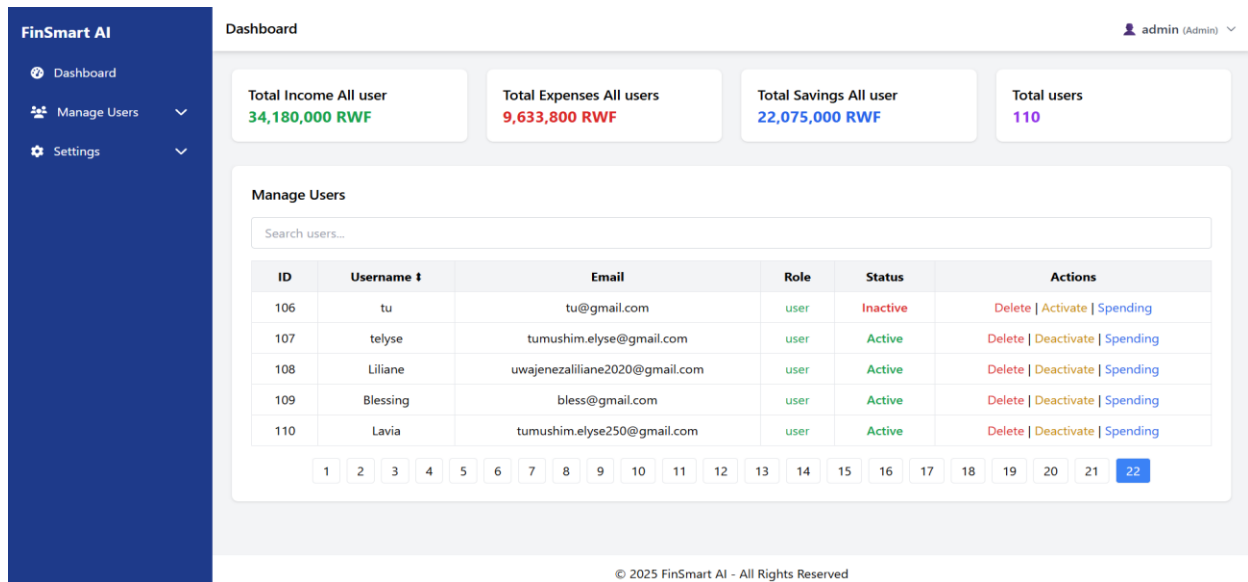


Figure 27: Admin dashboard

The admin can view a list of all users who have met their financial goals, showing their income, expenses, and savings status (view Figure 28)

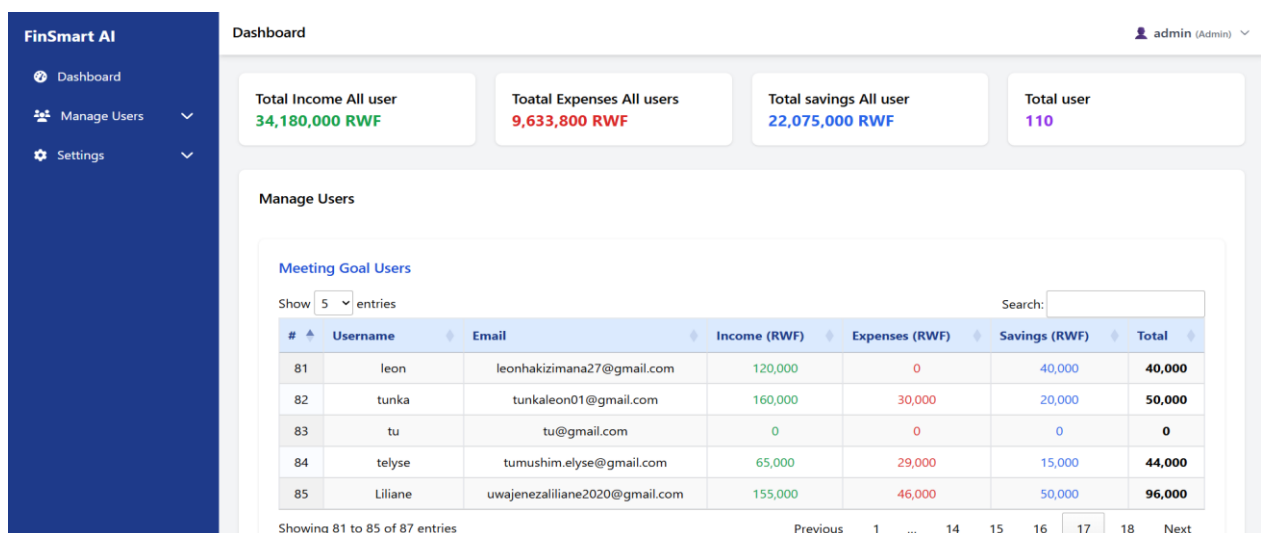


Figure 28: user who meet goal

The admin can view a list of users who have not met their financial goals, where income is less than the sum of expenses and saving goals.(look on Figure 29)

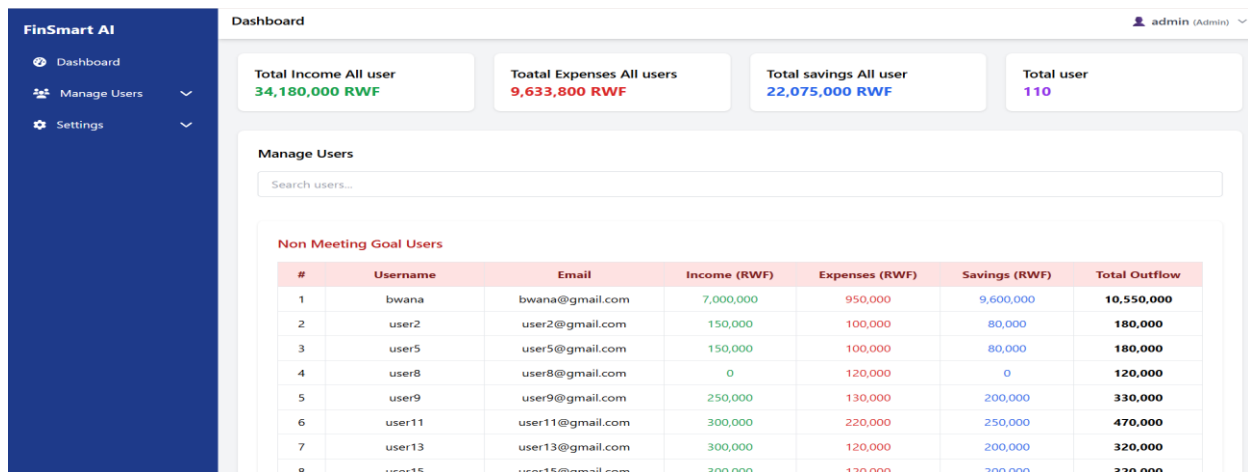


Figure 29:User who did not meet goal

The admin can view each user's spending dashboard, showing recent transactions with categories, amounts, and timestamps, providing a detailed overview of user activity (spending, recent, tracked) look on Figure 30



Figure 30: User spending dashboard

CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 CONCLUSIONS

The FinSmart AI project successfully met its primary objective of designing and developing an intelligent personal financial assistant that provides users with interactive, real-time financial guidance. The system features an AI-powered chatbot capable of offering budget suggestions, financial advice, and personalized insights based on users' spending habits and income records. This aligns directly with the goal of using machine learning to promote better financial decision-making and improve user engagement with personal finance tools.

The intuitive design of the system, combined with an accessible user interface, ensures ease of use even for individuals with minimal financial literacy or technical background. Preliminary testing showed a positive user response, indicating increased financial awareness, improved savings behaviour, and regular use of budgeting tools. Furthermore, the project's emphasis on privacy and ethical AI practices adds to its reliability and relevance. In conclusion, the system has demonstrated its effectiveness in empowering users to take control of their financial lives, and it offers a strong foundation for further development and deployment.

5.2 RECOMMENDATIONS

In light of the system's results and usability, the following recommendations are provided for individuals and institutions that may implement or benefit from the FinSmart AI system:

1. Encourage individual users to adopt FinSmart AI as a daily personal finance tool.

The system helps users track income, monitor expenses, and build smarter financial habits.

2. Support Kinyarwanda localization to improve accessibility.

Offering the system in Kinyarwanda will make it easier to use for all Rwandans, especially those with limited digital or language skill

3. Integrate mobile accessibility into FinSmart AI to increase user reach and convenience.

Ensuring the system works seamlessly on smartphones will enable more users, especially those in rural or low-connectivity areas, to manage their finances anytime, anywhere.

5.3 DISCUSSION AND KEY RESULTS

The implementation of FinSmart AI successfully demonstrates the ability of chatbots to support personal finance management. The system can record user data in real-time, analyze behavior, and provide advice. User testing showed that participants found the system helpful in tracking their money and understanding spending habits

5.4 CHALLENGES ENCOUNTERED

Training the Chatbot: Fine-tuning user intent recognition required multiple iterations to improve accuracy and relevance of responses.

Data Privacy Concerns: Ensuring the secure storage and handling of sensitive personal finance data was a critical priority throughout development.

Performance Issues: Optimizing the system to deliver fast response times during chatbot interactions was challenging, especially when processing complex queries and integrating with external APIs like OpenAI.

5.5 SUGGESTIONS FOR FURTHER STUDY

While the FinSmart AI system lays a solid foundation for intelligent financial assistance, further research and development are recommended to expand its functionality, adaptability, and long-term effectiveness:

1. Explore integration with mobile money platforms such as MTN Mobile Money and Airtel Money.

Future studies should examine how secure integration with mobile payment services can automate transaction tracking, improve data accuracy, and enhance real-time financial insights for users.

2. Investigate voice interaction and simplified navigation for non-digital users.

To increase usability for users with low literacy or limited digital experience, future development could explore localized voice commands and intuitive user interfaces.

3. Extend functionality to support small businesses and informal entrepreneurs.

Research could focus on adapting FinSmart AI to help entrepreneurs manage business income, expenses, and financial planning—bridging the gap between personal and business finance tools.

4. Develop ethical AI and data governance frameworks.

Further work is needed to ensure FinSmart AI operates with transparency, informed consent, and strong user data protections in line with ethical standards.

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APPENDICES

APPENDIX 1: WORK PLAN

No	TASK TO BE PERFORMED	DATE TO BE COMPLETED	PERSON ASSIGNED TO TASK
1	Project chosen and project proposal	On 22th Feb 2025-26th Feb 2025	TUMUSHIME Elyse
2	Project suggestion submission presentation	On 28th Feb 2025	TUMUSHIME Elyse
3	Data collection	On 05th Mar-25th Mar 2025	TUMUSHIME Elyse
4	Data analysis	On 26th Mar -20th Apr 2025	TUMUSHIME Elyse
5	Project design	On 21th Apr-20th June-2025	TUMUSHIME Elyse
6	Project testing	On 07th July- 24th June 2025	TUMUSHIME Elyse
7	Project submission	On 12 th July 2025	TUMUSHIME Elyse

Table 11: Work Plan

APPENDIX 2: GRANT CHART

No	START DATES	END DATES	TASKS	March	April	May	June
1	05/03/2025	20/04/2025	Research and data collection				
2	21/04/2025	24/06/2024	Implementation				
3	20/06/2024	10/07/2024	Writing report				

Table 12: grant chart

APPENDIX 3: PROJECT BUDGET

No	Items	Total Cost (RWF)
1	Preparation for the Study	25,000
2	The survey	25,000
3	Communication	50,000
4	Transport& Foods	200,000
5	Internet Connection	200,000
6	Documentation	Free
	TOTAL BUDGET	500,000

Table 13: Budget